

Electronic and Optoelectronic Devices based on Two-Dimensional Materials: From Fabrication to Application

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Since the rediscovery of graphene in 2004, this material has attracted an enormous amount of interest owing to its unique structural, mechanical, electronic, and optical properties. Beyond this, the unique properties of graphene have also triggered extensive research on other two-dimensional (2D) materials including transition metal dichalcogenides (TMDs), particularly for electronic and optoelectronic device applications. In particular, not only single junction but also various heterojunction devices based on 2D materials have afforded novel functionality and advances in many research fields. The availability of various 2D materials could allow the formation of a wide variety of heterojunctions with desired band alignments, thereby providing a powerful fabrication process platform for high-performance electronic and optoelectronic devices. In this review, the important fabrication techniques of i) doping, ii) contact engineering, and iii) heterojunction formation for improving the performance of 2D-material-based devices are first introduced. Promising 2D-material-based electronic and optoelectronic devices are then discussed, including lateral/vertical field-effect transistors (FETs), negative differential resistance (NDR) devices, memory, photodetectors, photovoltaics, and light-emitting diode (LED) devices.

promising and important 2D material. Graphene-based electronic and optoelectronic device applications^[11–15] have been widely researched to take advantage of its many outstanding properties, such as high carrier mobility ($10\,000\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ at room temperature),^[16] excellent optical transparency (97% for monolayer),^[17] high Young's modulus (0.5–1 TPa),^[18] and wide absorption spectrum (300–1400 nm).^[19]

The unique properties of graphene have also triggered extensive research in other 2D materials including TMDs, particularly for various electronic^[4,11–13,20–68] and optoelectronic device applications.^[3,5,14,15,24,31,69–103] 2D electronic devices have advanced from single junction to heterojunction devices and have recently been used in lateral/vertical field-effect transistors (FETs),^[4,11–13,20–41,62–64,67,68] negative differential resistance (NDR) devices,^[42–50,66] and memory devices.^[51–61,65] Especially, type-II heterojunctions have enabled efficient electron–hole separation during the

conversion from light to current, and these have found use in high-performance 2D optoelectronic devices such as photodetectors,^[3,5,14,15,24,31,69–83,102,103] photovoltaic devices,^[84–95] and light-emitting diodes (LEDs).^[87,88,95–101] The electronic and optoelectronic properties of semiconducting TMDs depend on the number of layers due to quantum confinement effects.^[104,105] For example, the transition from an indirect energy bandgap to a direct bandgap was observed for molybdenum- and tungsten-based TMDs as layer thickness decreases and approaches a monolayer. It has been reported that bulk molybdenum disulfide (MoS_2) and tungsten diselenide (WSe_2) have an indirect bandgap of 1.2 eV, whereas monolayer MoS_2 and WSe_2 have direct bandgaps of 1.88 eV and 1.65 eV, respectively, which lead to a high on/off-current ratio and high quantum efficiency.^[106] Meanwhile, rhenium disulfide (ReS_2) exhibited a direct bandgap of 1.44 eV independent of its layer thickness,^[107] and it is considered a promising candidate for future optoelectronic devices.^[24,31,33,82,83] In this review, we first introduce important fabrication techniques of i) doping, ii) contact engineering, and iii) heterojunction formation for improving the performance of 2D-material-based devices in terms of field-effect mobility, on-current, responsivity, and temporal response (Section 2). We then discuss promising 2D-material-based electronic (Section 3) and optoelectronic (Section 4) devices,

1. Introduction

Two-dimensional (2D) transition metal dichalcogenides (TMDs), which are sandwich structure compounds composed of a transition metal layer (Mo, W, etc.) and two chalcogen layers (S, Se, or Te), are an emerging family of layered materials that are held together by van der Waals interactions. These layered materials are cleaved into monolayer or few layer structures via mechanical^[1–6] or liquid phase exfoliation^[7–10] methods due to the weak van der Waals forces between neighboring layers. Since the discovery of graphene, a monolayer of carbon atoms bonded together in a hexagonal honeycomb lattice, in 2004, an enormous amount of interest has been given to it owing to its unique structural, mechanical, electronic, and optical properties. Graphene has rapidly become the most

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including lateral/vertical FETs, NDR devices, memory, photodetectors, photovoltaics, and LED devices.

2. Fabrication Technologies

2.1. Doping Process

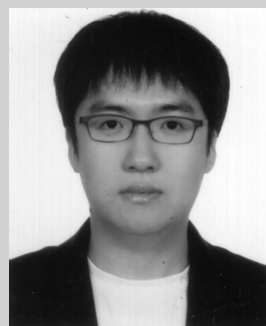
The development of doping techniques for 2D semiconductor materials is necessary in order to i) control the threshold voltage, ii) improve the field-effect mobility, iii) reduce the contact resistance between metal and semiconductor, iv) improve the responsivity, v) reduce the recombination of photoexcited carriers, vi) enhance the photoresponse, vii) convert the device type from n- to p- or p- to n-channel and/or viii) implement bandgap engineering. However, traditional doping processes like ion-implantation are not suitable for 2D materials due to crystal damage that occurs during implantation. Various doping techniques have recently been studied to improve the electrical and optical properties of 2D materials; these techniques include i) surface charge transfer doping,^[20,22,72,73,108–117] ii) interfacial dipole formation doping,^[74,118–123] iii) substitutional doping,^[124–127] and iv) fixed charge layer doping.^[128] These doping processes for 2D materials are summarized in Table 1.

The surface charge transfer phenomenon from dopant molecules to 2D materials mainly depends on the reduction potential of the dopant molecules and the chemical potential of the 2D materials. 1,1'-Dibenzyl-4,4'-bipyridinium dichloride (BV),^[108–111] nicotinamide adenine dinucleotide (NADH),^[112] polyethyleneimine (PEI),^[20] potassium (K),^[113] and triphenylphosphine (PPh₃)^[72,73] have a negative reduction potential, which resulted in the donation of electrons to 2D materials and a corresponding n-doping phenomenon in 2D materials such as graphene, MoS₂, WSe₂, and rhenium diselenide (ReSe₂) (Figure 1a). Cesium carbonate (Cs₂CO₃) was used to n-dope MoS₂ because the low work function of Cs₂CO₃ facilitated the transfer of electron carriers from Cs₂CO₃ to MoS₂.^[114] In contrast, 2,3-dichloro-5,6-dicyanobenzoquinone (DDQ),^[110] gold chloride (AuCl₃),^[109–111,115] 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F₄TCNQ),^[112,116] and 7,7,8,8-tetracyanoquinodimethane (TCNQ)^[112] dopants caused a p-type doping phenomenon in 2D materials due to their positive reduction potential. In addition, NO₂ has been used as a p-type dopant for 2D materials; in this case, electron carriers transferred from WSe₂ to NO₂ because of its strong oxidizing ability.^[22] A p-doping effect was also observed by depositing molybdenum trioxide (MoO₃) on MoS₂ because the Fermi level and the conduction band of MoO₃ were located below those of MoS₂, providing numerous electron carriers into the MoO₃ (Figure 1b).^[117] These surface charge transfer doping techniques were achieved by spin-coating (BV, DDQ, AuCl₃, and F₄TCNQ), drop-casting (NADH, TCNQ, and F₄TCNQ), gas flow (K, and NO₂), deposition (Cs₂CO₃, and MoO₃) on the 2D material samples, or immersion (AuCl₃, and PPh₃) of the samples in solution. In particular, MoS₂ and WSe₂ were doped to the degenerate level using BV, AuCl₃, K, and NO₂. Fang et al.^[22,113] reported high-performance MoS₂- and WSe₂-based FETs with high field-effect mobility by degenerately doping the source/drain (S/D) regions with K and NO₂ molecules, where the degenerately doped S/D



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regions experienced a reduction in S/D contact resistance. Jo et al.^[72,73] also reported high-performance 2D material (MoS₂, WSe₂, and ReSe₂)-based transistors and photodetectors by improving the source-side contact resistance through a PPh₃-based n-doping technique. Furthermore, Cs₂CO₃ was found to induce n-type doping in MoS₂, and an enhancement in

Table 1. Doping processes of 2D materials.

Material	Dopant	Doping scheme	Doping type	Doping concentration [cm ⁻²]	Stability	Ref.
Graphene	BV ^{a)}	Surface charge transfer	n-type	10 ¹³	–	[110]
	N	Substitution		–	–	[126]
	DDQ ^{b)}	Surface charge transfer	p-type	10 ¹³	–	[110]
	AuCl ₃			10 ¹³	–	[110,115]
	Ga, Ge, As, Se	Substitution	–	–	–	[127]
MoS ₂	BV		n-type	10 ¹³	>9 days	[108,109]
	NADH ^{c)}			–	–	[112]
	PEI ^{d)}			–	<6 hours	[20]
	K			10 ¹³	Unstable	[113]
	PPh ₃ ^{e)}			10 ¹¹	>14 days	[73]
	Cs ₂ CO ₃			10 ¹²	>3 days	[114]
	AuCl ₃		p-type	10 ¹³	–	[109,111]
	TCNQ ^{f)}			–	–	[112]
	F ₄ TCNQ ^{g)}			–	–	[112]
	MoO ₃			–	–	[117]
	SAM ^{h)} (-NH ₂ , -SH)	Dipole	n-type	10 ¹¹ –10 ¹²	<6 hours	[74,118,119]
	DNA ⁱ⁾			10 ¹⁰	<3 days	[121]
	Ln-DNA (Eu ³⁺ , Gd ³⁺ , Tb ³⁺ , Er ³⁺)			10 ¹⁰	>5 days	[122]
	PSG ^{j)}			10 ¹⁰ –10 ¹³	>4 days	[123]
	SAM (-CH ₃)		p-type	10 ¹¹ –10 ¹²	<6 hours	[74,118–120]
	M-DNA (Zn ²⁺ , Ni ²⁺ , Co ²⁺ , Cu ²⁺)			10 ¹¹	>8 days	[121]
	Nb	Substitution	p-type	10 ¹⁴	> month	[124,125]
WSe ₂	K	Surface charge transfer	n-type	10 ¹²	Unstable	[113]
	PPh ₃			10 ¹¹	>5 days	[72]
	F ₄ TCNQ		p-type	–	>2 weeks	[116]
	NO ₂			10 ¹³	Unstable	[22]
	SAM (-NH ₂)	Dipole	n-type	10 ¹¹	<6 hours	[74]
	DNA			10 ¹⁰	<3 days	[121]
	Ln-DNA (Eu ³⁺ , Gd ³⁺ , Tb ³⁺ , Er ³⁺)			10 ¹⁰	>5 days	[122]
	SAM (-CH ₃)		p-type	10 ¹¹	<6 hours	[74,120]
	M-DNA (Zn ²⁺ , Ni ²⁺ , Co ²⁺ , Cu ²⁺)			10 ¹¹	>8 days	[121]
	SiN _x	Fixed charge		10 ¹⁰ –10 ¹³	–	[128]
ReSe ₂	PPh ₃	Surface charge transfer	n-type	10 ¹¹	>14 days	[73]

^{a)} BV, 1,1'-dibenzyl-4,4'-bipyridinium dichloride; ^{b)} DDQ, 2,3-dichloro-5,6-dicyanobenzoquinone; ^{c)} NADH, nicotinamide adenine dinucleotide; ^{d)} PEI, polyethyleneimine; ^{e)} PPh₃, triphenylphosphine; ^{f)} TCNQ, 7,7,8,8-tetracyanoquinodimethane; ^{g)} F₄TCNQ, 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane; ^{h)} SAM, self-assembled monolayers; ⁱ⁾ DNA, deoxyribonucleic acid; ^{j)} PSG, phosphorus silicate glass.

photocurrent was observed in Cs₂CO₃-deposited MoS₂-based photodetectors where supplied electrons were expected to increase the formation of tightly bounded negative trions in MoS₂ and reduce the charge recombination of photocarriers.^[114]

Doping of 2D materials has also been achieved by the formation of molecular dipole moments.^[74,118–123] Self-assembled monolayers (SAMs) have attracted attention for controllable doping of 2D materials because the dipole moments and polarities of SAMs can be easily tuned by selecting various functional groups. Since amine (-NH₂), fluoro (-CF₃), and thiol (-SH) functional groups show negative polarities, the

electron carriers are pushed into the FET channel from the interface of the SAM and 2D material, leading to n-doping of MoS₂ and WSe₂.^[74,118,119] In contrast, methyl (-CH₃) functional groups with positive polarity were utilized to p-dope MoS₂ and WSe₂.^[74,118–120] Deoxyribonucleic acid (DNA) nanostructures were also used to dope 2D materials. DNA and Ln-DNA (with trivalent Ln ions of Eu³⁺, Gd³⁺, Tb³⁺, Er³⁺) nanostructures showed negative polarity, resulting in n-doped 2D materials (Figure 1c).^[121,122] Conversely, M-DNA nanostructures with positive metal ions (divalent metal ions of Zn²⁺, Ni²⁺, Co²⁺, and Cu²⁺) showed positive polarity and

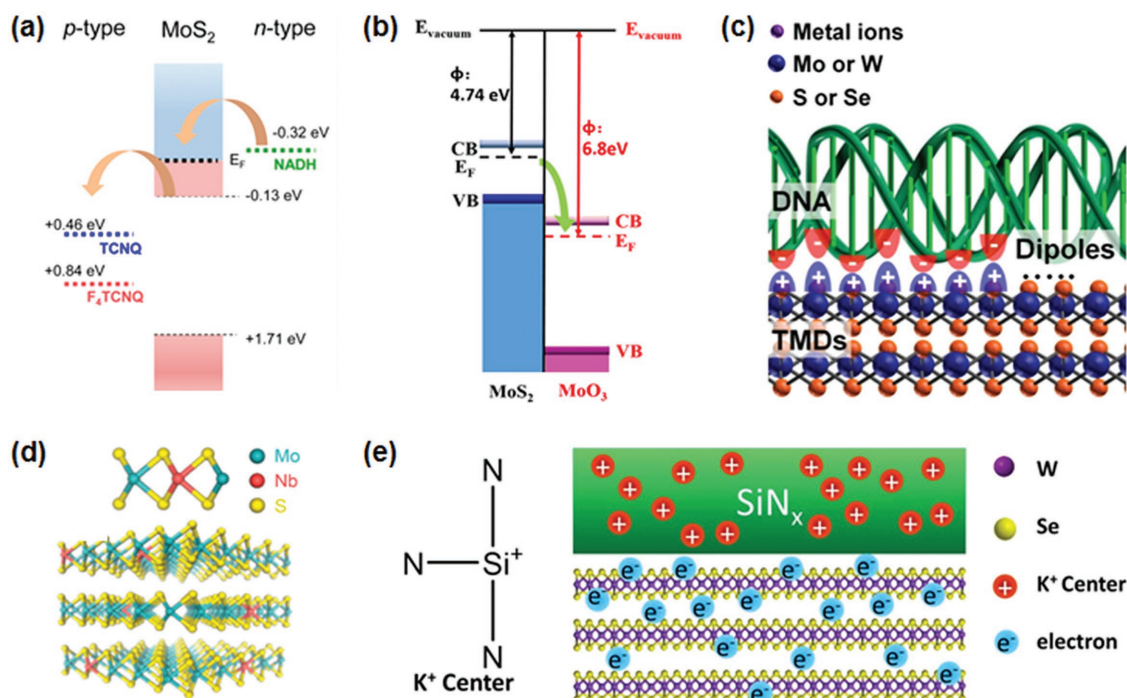


Figure 1. a) Schematic of relative potentials for monolayer MoS₂, n-type dopant (NADH), and p-type dopants (TCNQ and F₄TCNQ), which explains the surface charge transfer doping mechanism. Reproduced with permission.^[112] Copyright 2013, American Chemical Society. b) Energy band diagram for MoS₂ and MoO₃ explaining charge transfer from MoS₂ to MoO₃. Reproduced with permission.^[117] Copyright 2013, AIP Publishing LLC. c) Schematic diagram of DNA-treated TMD film showing n-type doping of TMD due to the negative polarity of DNA. Reproduced with permission.^[121] Copyright 2014, American Chemical Society. d) Schematic illustration of MoS₂:Nb layers, where Nb dopants were substitutionally incorporated into MoS₂ by replacing the Mo atoms. Reproduced with permission.^[124] Copyright 2014, American Chemical Society. e) Schematic of SiN_x structure with a positively-charged K⁺ center, and the schematic diagram of SiN_x-deposited WSe₂. The positively-charged K⁺ centers attract electron carriers inside the WSe₂. Reproduced with permission.^[128] Copyright 2014, AIP Publishing LLC.

reduced the electron carrier density in 2D materials, resulting in p-doping.^[121] Threshold voltage and field-effect mobility of 2D-material-based FETs were modified using SAM and DNA doping techniques; these doping processes were performed by immersing device samples in solutions containing the SAM molecules or by drop-casting DNA solutions onto the samples.^[74,118–122] In addition, the recombination probability of excitons was decreased by increasing the number of tightly bound triions in –NH₂ functionalized and DNA/Ln-DNA-doped MoS₂.^[74,121,122] The recombination probability of negatively-charged triions was suppressed by decreasing the electron concentration in –CH₃-functionalized and M-DNA-doped WSe₂.^[74,120,121] This consequently increased the photocurrents of photodetectors fabricated on n-doped MoS₂ and p-doped WSe₂. In particular, a wide-range controllable n-doping methods (from nondegenerate to degenerate) was reported by Park et al.,^[123] where a phosphorus silicate glass (PSG) insulating layer was used to modulate the electron carrier density of MoS₂. This method consisted of three steps: i) dopant out-diffusion, ii) thermal activation, and iii) optical activation processes. Negatively-charged oxygen (O) atoms of phosphorus pentoxide (P₂O₅) molecules in the PSG layer were expected to attract hole carriers from the MoS₂ layer to the MoS₂/PSG interface, resulting in n-doping of the MoS₂ layer. The n-doping range was exceptionally broad, from 3.6×10^{10} to $8.3 \times 10^{12} \text{ cm}^{-2}$.

Doping techniques based on surface charge transfer and interfacial dipole formation are normally unstable in ambient air. To achieve more stable doping, a substitutional doping process using niobium (Nb) was applied to MoS₂.^[124,125] Nb has one less valence electron than Mo and is therefore a suitable substitutional acceptor. Nb dopants substitute for Mo atoms in MoS₂ and remove an electron carrier from the valence band of MoS₂, leading to a p-type doping effect (Figure 1d). The substitutional Nb doping of MoS₂ was also stable for more than one month in an air ambient due to covalently bonded Nb atoms inside the lattice.^[124] Another stable doping technique for 2D materials is deposition of a fixed charge layer onto the 2D materials. Chen et al.^[128] achieved stable n-doping of WSe₂ by depositing silicon nitride (SiN_x) on a WSe₂ surface, and they demonstrated a WSe₂-based n-FET with high electron mobility. This method was effective because SiN_x had positive fixed charge centers, which originated from ⁺Si≡N₃ dangling bonds known as K⁺ centers (Figure 1e). Thus, an n-doping phenomenon was observed in the WSe₂ sample with SiN_x.^[128] The electron carrier concentration in WSe₂ was controlled up to the degenerate level (7.4×10^{11} and $9.5 \times 10^{13} \text{ cm}^{-2}$) by adjusting the stoichiometry of the SiN_x via gas flow control. Meanwhile, C atoms in the graphene lattice can be replaced by N, Ga, Ge, As, and Se atoms, subsequently forming covalent bonding with C atoms. This suppressed the density of states of graphene near its Fermi

level and it consequently opened the gap between conduction and valence bands in the range of 0.05–0.04 eV.^[126,127]

The surface charge transfer doping technique on 2D materials was capable of achieving nondegenerate doping levels using PPh₃^[72,73] and Cs₂CO₃^[114] as well as degenerate doping levels using BV^[108–110] DDQ^[110] AuCl₃^[109–111,115] K^[113] and NO₂.^[22] In particular, the BV^[108,109] and PPh₃-doped MoS₂^[73] exhibited excellent electrical stability even after being exposed in ambient air for 9 days and 14 days, respectively. On the other hand, because K atoms are highly reactive in air, K-doped MoS₂ and WSe₂ were demonstrated in vacuum. Meanwhile, most of the interface dipole formation doping techniques resulted in a low-level doping concentration (nondegenerate regime) for 2D materials with a moderate stability in ambient air (3–8 days).^[74,118–123] In the case of substitutional doping processes, such as the Nb-doped MoS₂, superior air-stability (more than 1 month) was observed because the substitutional dopants (Nb) were covalently bonded inside the lattice.^[124] Excellent doping stability via depositing an insulator containing fixed charges such as SiN_x on 2D materials was also reported.^[128]

2.2. Contact Engineering

Low contact resistance (R_c) is also an important factor to achieve high device performance (e.g., high on-current^[4] or large photoresponse^[3]) in 2D-semiconductor-based electronics and optoelectronics. However, a considerable contact resistance between metals and 2D materials, which is caused by the van der Waals gap,^[129] prevents the successful fabrication of various 2D-material-based devices, such as FETs,^[23,130–136] photodetectors,^[3,137,138] photovoltaic devices,^[84,92,139] and LEDs.^[96–98] Many studies have recently focused on reducing the contact resistance of 2D-material-based devices through diverse strategies: i) an increase in edge-contacted perimeter length,^[13,137,140,141] ii) insertion of an insulator or a 2D layer at the metal/2D material interface,^[138,142–145] iii) usage of degenerate doping,^[113] or iv) application of phase engineering^[146] for contacts (i.e., S/D regions in FET) of 2D materials. The electrical characteristics of these metal/2D material contacts are summarized in **Table 2**.

Figure 2a shows two types of contact geometries (top and edge contacts) between the metal and 2D material. A top contact

Table 2. Electrical characteristics of metal/2D material contacts.

Material	Contacted electrode	Method	Schottky barrier height [eV]	Contact resistance [$\Omega \mu\text{m}$]	Ref.
Graphene	Ti	–	–	750	[140]
	Ni	–	–	250	[130]
	Ti	Edge contact	–	400	[140]
	Cr	Edge contact	–	140	[13]
	Ti	Edge contact + Doping	–	23	[137]
MoS ₂	Ni	–	–	4700	[131]
	Mo	–	–	2000	[132]
	Graphene	–	0.022	–	[133]
	Sc	–	0.030	–	[21]
	Ti	–	0.065	–	[148]
	Co	–	0.121	–	[142]
	Au	–	0.150	–	[149]
	Pt	–	0.230	–	[21]
	TiO ₂ /Co	Inserting layer	0.027	–	[142]
	MgO/Co	Inserting layer	0.0097	–	[143]
	Au	–	–	1100	[146]
	1T-MoS ₂ /Au	Phase change	–	240	[146]
WS ₂	Cr	–	–	31 300	[134]
	Cr	Doping	–	930	[134]
WSe ₂	Graphene	–	–	2000	[23]
	Pd	–	0.326	–	[153]
	NbSe ₂	–	0.199	–	[153]
	Nb _{1-x} W _x Se ₂ /NbSe ₂	Doping	0.176	–	[153]
	Nb _{0.005} W _{0.995} Se ₂ /Ti	Doping	–	300	[154]
BP	Ni	–	–	3150	[135]
	Pd	–	–	1750	[135]
	Ti	–	–	1140	[136]

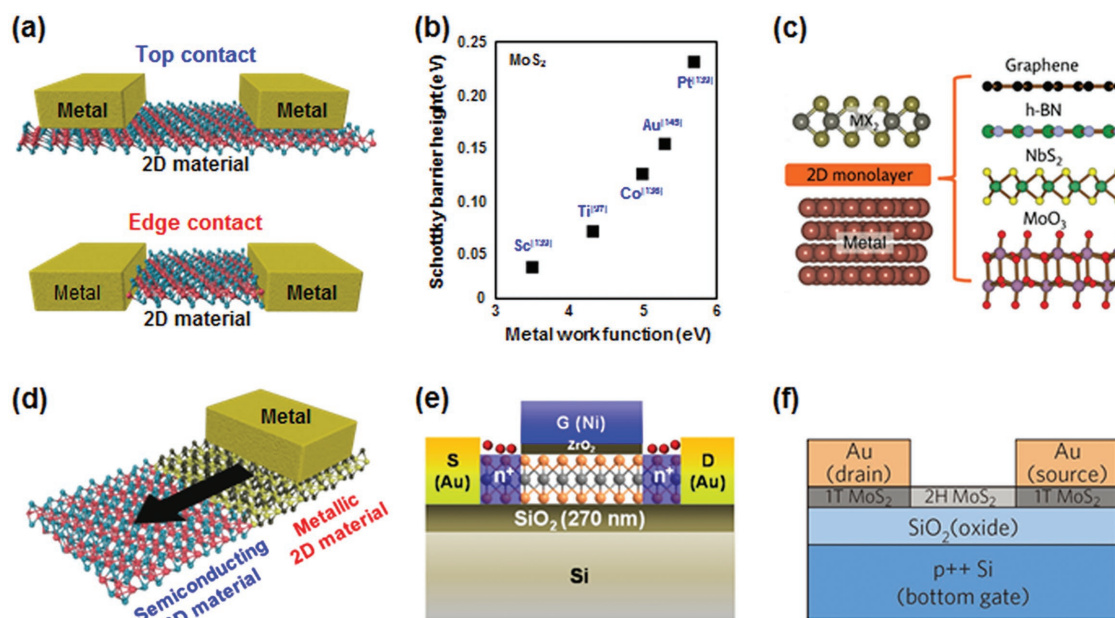


Figure 2. a) Schematic diagrams of two types of contact geometries (top and edge contacts) between metals and 2D materials. b) Schottky barrier height (Φ_B) values for previously reported MoS_2 devices as a function of metal work function (Φ_m). c) Cross-sectional diagram of metal/TMD interface with insertion of various 2D layers. Reproduced with permission.^[145] d) Contact scheme between metallic and semiconducting 2D materials. e) WSe_2 -based FET with degenerately potassium-doped contact sides (S/D regions). Reproduced with permission.^[113] Copyright 2013, American Chemical Society. f) MoS_2 -based FET with metallic 1T- MoS_2 contact by selective phase engineering (from 2H to 1T phase). Reproduced with permission.^[146] Copyright 2014, Nature Publishing Group.

was easily formed by contacting the metal onto the top surface of the 2D material. However, because of the very smooth top surface of 2D materials (without dangling bonds), the interface between the metal and 2D material is only bonded by weak van der Waals forces, and this interface gap forms an additional tunneling barrier at the metal/2D material junction.^[129] This gap decreases the charge injection probability from the metal to the 2D material, resulting in a higher contact resistance.^[129] In the case of edge contact, stronger orbital overlap was observed at the metal/2D material interface, and a lower tunneling barrier was obtained compared to the top contact.^[141] In a previous study on metal-graphene contact resistance, Park et al.^[137] reported a significant reduction in contact resistance as a function of the edge-contacted perimeter length; they were successful in minimizing the R_c to $23 \, \Omega \, \mu\text{m}$ and improving the performance (responsivity and detectivity) of a graphene-based photodetector. Even though the edge contact showed higher conductivity, the optimization of the top contact is still important because i) the top surface usually has a much larger area than the edge side, ii) the formation of pure edge contact (without overlap of top surface) is very difficult in atomically thin 2D materials, and iii) there are relatively few studies related to edge contact in other 2D materials.^[13,137,140,141] In light of this, further studies on both the top and edge contacts are needed to resolve the contact issues in various 2D materials.

Because the carrier transport from metals to 2D materials is strongly dependent on the Schottky barrier height (Φ_B) at the junction interface, precise control of Φ_B is very important for optimizing 2D-material-based devices. Ideally, Φ_B is determined by the difference between the work function (Φ_m) of the metal and the electron affinity (χ) of the 2D material (Schottky–Mott

rule).^[147] However, the metal Fermi level at the metal/2D interface is usually pinned at a specific energy level, which makes it difficult to control Φ_B by changing the contact metal. Figure 2b presents the Φ_B values between MoS_2 and various metals as a function of Φ_m .^[21,142,148,149] The Fermi levels seem to be pinned 0.05 to 0.20 eV below the conduction band minimum (CBM), indicating that the Φ_B values are only weakly dependent on Φ_m . In order to resolve this pinning problem, many studies related to depinning of the Fermi level were recently conducted. These studies focused on inserting insulating^[142,143] or 2D layers (graphene,^[144] hexagonal boron nitride (h-BN),^[138] or TMD^[145]) between the metals and the 2D materials. As shown in Figure 2c, Farmanbar et al.^[145] theoretically demonstrated a significant reduction in Schottky barrier height and an improvement in contact resistance through the insertion of various 2D layers at the metal/TMD interface. In addition, enhanced optoelectronic characteristics (an increase in photocurrent) were also observed after inserting an h-BN (2D insulating layer) at the metal/2D interface.^[138]

Apart from manipulation of metal/2D junctions, one of the most promising strategies to reduce contact resistance in 2D-material-based devices is to modify the electrical properties of the S/D regions (from semiconducting to metallic). As shown in Figure 2d, native chemical bonds at the interface between semiconducting and metallic materials were expected to result in a higher contact conductivity compared to the direct contact of three-dimensional (3D) metals onto semiconducting 2D materials.^[21,142,148,149] Related theoretical^[150–152] and experimental^[113,146,153,154] results obtained in recent studies showed enhancements in contact resistance and device performance. Degenerate doping^[113] and phase engineering^[146] have

recently been used to form a metallic region in the semiconducting 2D materials. A dramatic increase in on-current was observed in a WSe₂-based FET after a decrease in contact resistance was realized after degenerate doping in S/D regions (Figure 2e).^[113] Furthermore, as shown in Figure 2f, a junction contact consisting of metallic 1T-MoS₂ (S/D regions) and semiconducting 2H-MoS₂ (channel region) was achieved by using selective phase engineering (changing phase from 2H to 1T).^[146] After the phase change process, an extremely low contact resistance (240 Ω μm) was obtained, which improved the electronic device performance.

2.3. Heterojunction Formation

Van der Waals heterojunction-based electronics and optoelectronics have recently received considerable attention due to i) the atomically thin structure of the junction, which provides novel functionalities (e.g., gate bias tunable rectifying behavior and photovoltaic response), ii) the absence of dangling bonds on the surface, which allows a high-quality heterojunction interface without any interfacial defect problems caused by a lattice constant difference, and iii) ultra-fast charge transfer, which dramatically enhances the temporal response. Furthermore, unique electrical and optical junction properties can be achieved by combining various 2D materials with different work functions, band gaps, and electron affinities. Indeed, various 2D heterojunction-based devices have recently been reported including transistors,^[34–41] memory,^[57,58,61] NDR devices,^[42,45,47–50,66] photodetectors,^[77,78,86,94,102,103,155–157] photovoltaic devices,^[89,91,158] and LEDs.^[100,101] Here, we introduce the recent research trend of electronic and optoelectronic devices based on 2D heterojunctions, which are summarized in Table 3.

Table 3. Electronic and optoelectronic devices based on 2D heterojunction.

Materials	Application	Method	Properties ^{a)}	Ref.
Gr ^{b)} /h-BN/Gr	FET ^{c)}	Dry transfer	On/off-current ratio: 10 ⁴	[34]
Gr/MoS ₂			On/off-current ratio: 10 ³	[35]
Gr/WSe ₂ /Gr			On/off-current ratio: 10 ⁶	[40]
Gr/MoS ₂			On/off-current ratio: 10 ⁵	[37]
Gr/MoS ₂			On/off-current ratio: 5 × 10 ⁴	[38]
Gr/MoS ₂			On/off-current ratio: 10 ²	[36]
Gr/WSe ₂			On/off-current ratio: 10 ⁴	[41]
Gr/MoS ₂	Memory		P/E ^{d)} ratio: 10 ⁴ , Endurance: >120	[57]
Gr/h-BN/MoS ₂			P/E ratio: 10 ⁵ , Endurance: >110	[58]
BP/h-BN/MoS ₂			P/E ratio: 10 ²	[61]
Gr/h-BN/Gr	NDR ^{e)} device		PVCR ^{f)} : 4 at 7 K	[42]
Gr/h-BN/Gr			Multiple NDR peaks	[45]
WSe ₂ /MoSe ₂ /Gr		Direct growth	PVCR: 1.9	[47]
MoS ₂ /WSe ₂ /Gr			PVCR: 2.2	[47]
MoS ₂ /WSe ₂		Dry transfer	PVCR: 1.86 at 77 K	[48]
MoS ₂ /WSe ₂			PVCR: 1.6	[49]
BP/SnS ₂			PVCR: 1.8	[50]
MoS ₂ /WSe ₂			On/off-current ratio: 10 ⁴	[66]
MoS ₂ /WS ₂	Photodetector		T ^{g)} : 50 fs	[155]
MoS ₂ /WS ₂			T: 50fs	[102]
Gr/WSe ₂ /Gr			R ^{h)} : 20mA/W, EQE ⁱ⁾ : 7.3%, T: 5.5 ps	[77]
MoS ₂ /WSe ₂			R: 2mA W ⁻¹ , EQE: 34%	[94]
GaTe/MoS ₂			R: 1.3 A W ⁻¹ , EQE: 266%	[156]
Gr/WS ₂ /Gr			R: 0.1 A W ⁻¹ , EQE: 30%	[157]
Gr/MoS ₂ /Gr			R: 0.22 A W ⁻¹ , EQE: 55%	[86]
rGO/MoS ₂			R: 2.4 × 10 ⁴ A W ⁻¹	[103]
MoS ₂ /Gr/WSe ₂			R: 10 ⁴ A W ⁻¹ , EQE: 10 ⁶ %	[78]
GaTe/MoS ₂	Photovoltaic device		EQE: 61.68%, PCE ^{j)} : 0.45%	[91]
ReSe ₂ /MoS ₂			EQE: 1266%, PCE: 0.072%	[158]
MoS ₂ /WSe ₂			EQE: 1.5%, PCE: 0.2%	[89]
Gr/h-BN/TMDs ^{k)} /h-BN/Gr	LED ^{l)}		–	[100,101]

^{a)} Properties at room temperature; ^{b)} Gr, graphene; ^{c)} FET, field-effect transistor; ^{d)} P/E, program/erase; ^{e)} NDR, negative differential resistance; ^{f)} PVCR, peak-to-valley current ratio; ^{g)} T, response time; ^{h)} R, responsivity; ⁱ⁾ EQE, external quantum efficiency; ^{j)} PCE, power conversion efficiency; ^{k)} MoS₂, MoSe₂, WS₂, and WSe₂; ^{l)} LED, light-emitting diode.

2.3.1. 2D Heterojunction-Based Electronics

The zero band gap property of graphene makes it difficult to achieve high on/off-current ratios in graphene transistors. Therefore, various graphene-based heterojunction devices have recently been demonstrated by vertically stacking another 2D material (h-BN, MoS₂, WSe₂, and so on) onto the graphene layer.^[34–41] Although graphene has metallic properties, its Fermi level can be modulated by an applied electric field, which allows for adjustment of the Schottky barrier height between graphene and another 2D material. Britnell et al.^[34] first introduced a graphene-based 2D heterojunction vertical transistor exploiting

this interesting semi-metallic property of graphene; in this work, the on/off-current ratio was about 50 and 10⁴ for h-BN and MoS₂ devices, respectively (Figure 3a). In addition, very high on/off-current ratio values between 10³ and 10⁶ have been achieved at room temperature under a low operation bias (V_{DS} = 0.1 V) in graphene-based 2D heterojunction transistors with various 2D materials such as MoS₂, WS₂, and WSe₂.^[35–41] Yu et al.^[35] demonstrated a complementary inverter by sequentially stacking graphene, Bi₂Sr₂Co₂O₈ (as a p-channel), graphene, and MoS₂ (as a n-channel) on a SiN_x/Si substrate. Georgiou et al.^[40] also showed the possibility of flexible and transparent electronics by transferring a graphene/WS₂ heterojunction

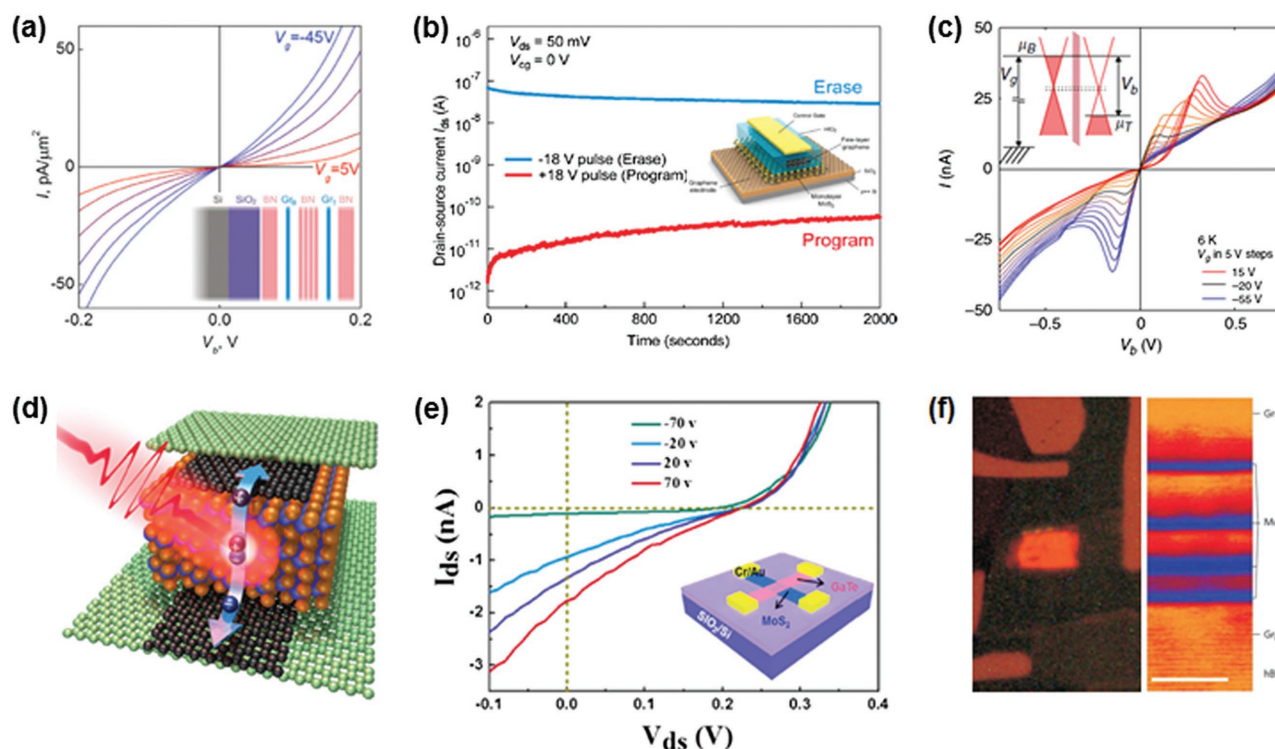


Figure 3. a) I_D - V_D characteristics of a graphene/h-BN/graphene heterojunction-based vertical transistor at different gate voltages. The inset shows a schematic of the graphene/h-BN/graphene vertical transistor. Reproduced with permission.^[34] Copyright 2012, American Association for the Advancement of Science. b) Retention performance of a MoS_2 /graphene heterostructure memory device, where V_{DS} is 50 mV, and the duration of the control-gate voltage pulse is 3 s. The inset presents a three-dimensional schematic illustration of the MoS_2 /graphene heterojunction-based memory device. Reproduced with permission.^[57] Copyright 2013, American Chemical Society. c) I - V characteristics of graphene/h-BN/graphene heterojunction-based RTD at 6 K. The inset shows a schematic band diagram of the graphene/h-BN/graphene heterostructure under forward bias with $V_{GS} = 15$ V. Reproduced with permission.^[42] Copyright 2013, Macmillan Publishers Limited. d) Schematic of h-BN/graphene/ WSe_2 /graphene/h-BN heterostructure. Photoexcited electron and hole carriers were separated into opposite directions. Reproduced with permission.^[77] Copyright 2016, Macmillan Publishers Limited. e) I - V characteristics of GaTe/ MoS_2 heterojunction-based photovoltaic device at various gate voltages. The inset shows a schematic of GaTe/ MoS_2 heterostructure. Reproduced with permission.^[91] Copyright 2015, American Chemical Society. f) EL optical (left) and cross-sectional bright-field SEM images (right) of a MoS_2 multiple-QW heterostructure. Reproduced with permission.^[100] Copyright 2015, Macmillan Publishers Limited.

transistor onto a polyethylene terephthalate (PET) film. These results showed that graphene-based heterojunction vertical transistors are viable options for future low-power and flexible electronic devices.

Heterojunction structures based on 2D materials have also been used for memory applications, which are important building blocks in digital electronics.^[57,58,61] Bertolazzi et al.^[57] demonstrated nonvolatile memory devices with a MoS_2 /graphene heterostructure, where hafnium oxide (HfO_2), MoS_2 , and graphene were used as the tunneling oxide, channel, and charge trapping layers, respectively (Figure 3b). This device could be switched between the program and erase states by transferring electron carriers from the channel (MoS_2) to the floating gate (graphene) or vice versa through the barrier layer (HfO_2). Similarly, graphene/h-BN/ MoS_2 heterojunction-based memories with high erase/program ratios of 10^5 were also reported by Choi et al.^[58] Furthermore, Li et al.^[61] also showed nonvolatile memory devices made by stacking black phosphorus (BP)/h-BN/ MoS_2 heterostructures on a SiO_2/Si substrate, where BP, h-BN, and MoS_2 were employed as channel, barrier, and charge trapping layers, respectively.

Tunneling diodes with NDR characteristics for future low-power applications represent another promising application

based on 2D heterojunctions. Because the NDR characteristic provides multiple threshold voltage values, this tunneling device is expected to be used to realize multivalued logic (MVL) circuits that require a higher number of logic states. Recently, NDR properties were achieved through resonant tunneling of electron carriers in graphene/h-BN/graphene (Figure 3c), WSe_2 /molybdenum diselenide (MoSe_2)/graphene, and MoS_2 / WSe_2 /graphene heterostructures.^[42,45,47] This resonance occurs when electronic spectra of two layers separated by a thin barrier layer are aligned. MoS_2 / WSe_2 and BP/tin diselenide (SnSe_2) hetero-PN-junction devices based on an Esaki diode scheme have also exhibited NDR behavior. This was attributed to the band-to-band tunneling and diffusion of electron carriers between the conduction band of one layer and the valence band of another layer.^[48–50] Roy et al.^[48] demonstrated MoS_2 / WSe_2 heterojunction-based NDR devices. In their work, carrier concentrations of MoS_2 and WSe_2 were independently controlled by two separate gate electrodes. A ternary inverter with three stable logical states was fabricated as a representative MVL application by using a MoS_2 / WSe_2 heterojunction device.^[49] A heterojunction device composed of SnSe_2 and BP was also reported to have NDR behavior with a peak-to-valley current

ratio (PVCR) of 1.8 at room temperature.^[50] Meanwhile, Huo et al. observed a sharp peak in the I_D - V_G curve of a $\text{MoS}_2/\text{WSe}_2$ heterojunction-based transistor.^[66] When a large negative gate and drain biases were applied, large amounts of hole carriers were tunneled from WSe_2 into MoS_2 , reducing the drain current (from MoS_2 to WSe_2) and consequently forming a sharp current peak.

2.3.2. 2D Heterojunction-Based Optoelectronics

2D-material-based heterojunctions, which can be easily obtained by stacking p- and n-type 2D materials, have been applied to various optoelectronic devices, such as photodetectors, photovoltaic devices, and LEDs. Graphene/ WSe_2 /graphene (Figure 3d) and MoS_2/WS_2 heterostructures exhibited very efficient electron-hole separations with fast photoresponse times of 5.5 ps^[77] and 50 fs,^[102,155] respectively. After photoexcitation, electron-hole pairs and interlayer hot excitons with excess energy were generated in each layer. Then, the electron-hole pairs were quickly separated by the excess energy of the interlayer hot excitons.^[102] In addition, a gate-tunable photovoltaic response was also reported in $\text{MoS}_2/\text{WSe}_2$ ^[94] and gallium telluride (GaTe)/ MoS_2 ^[156] heterojunction photodetectors, where the amplitude and polarity of the photocurrent were modulated by electrostatic gating. Yu et al.^[86] and Britnell et al.^[157] reported highly efficient photocurrent generation in vertical heterostructures consisting of 2D materials (graphene/ MoS_2 /graphene and graphene/ WS_2 /graphene, respectively). In these works, the graphene layers served as transparent and tunable electrodes, and charge separation and transport could be directly modulated by the external electric field across the gate electrode because the field direction was parallel to the carrier transport direction.^[86,157] A reduced graphene oxide (rGO)/ MoS_2 heterojunction-based photodetector was also reported by Yang et al.^[103] The recombination rate of photoexcited electrons and holes was suppressed due to a built-in potential at the rGO/ MoS_2 interface, leading to high photoresponsivity ($2.4 \times 10^4 \text{ A W}^{-1}$) in the rGO/ MoS_2 photodetector. Furthermore, Long et al.^[78] demonstrated a broadband MoS_2 /graphene/ WSe_2 heterojunction photodetector that was able to absorb wide spectral light (from 400 nm to 2400 nm) because gapless graphene has a unique property (hot-electron photothermoelectric detection) that induces strong interband absorption over a broad range of wavelengths.^[159] A GaTe/MoS_2 heterostructure device with a high photodetection efficiency (external quantum efficiency (EQE): 61.68%) and photovoltaic (power conversion efficiency (PCE): 0.45%) performance was demonstrated by Wang et al.,^[91] where the GaTe has a direct bandgap independent of its thickness (Figure 3e). Similarly, $\text{ReSe}_2/\text{MoS}_2$ ^[158] and $\text{WSe}_2/\text{MoS}_2$ ^[89] heterostructure devices also exhibited a photovoltaic effect with PCE values of 0.072% and 0.2%, respectively. Quantum well (QW) structures achieved by stacking h -BN and TMD materials were used to develop LEDs, where electron and hole carriers were recombined in the QWs, eventually emitting a photon.^[100,101] As shown in Figure 3f, Withers et al.^[100,101] demonstrated LEDs by stacking graphene, h -BN, and various TMD materials (MoS_2 , MoSe_2 , WSe_2 , and WS_2). The quantum efficiency (QE) was dramatically enhanced (10%) compared to that of a single

QW structure (graphene/ h -BN/ MoS_2 / h -BN/graphene) (1%) by using multiple QW structures (graphene/ h -BN/ MoS_2 / h -BN/ MoS_2 / h -BN/ MoS_2 / h -BN/graphene).^[100]

3. Application of 2D Materials to Electronics

Monolayer 2D materials (e.g., graphene, MoS_2 , WSe_2) are expected to be suitable for the channel materials in future high-performance electronic devices due to their thickness scalability at the atomic level (about 0.7 nm). Generally, an ultra-thin semiconductor can be used to enhance the gate controllability of devices because the source-to-drain current path is extremely close to the gate electrode.^[160,161] In addition, monolayer 2D materials are stable under high mechanical strain.^[18,162] Therefore, they are being considered as promising candidates for channel materials for flexible and stretchable electronic applications. These unique properties indicate the feasibility of 2D materials for use in next-generation electronic devices. In this section, we review electronic devices based on 2D materials: i) FETs, ii) NDR devices, and iii) memory devices.

3.1. Field-Effect Transistor (FET)

3.1.1. Lateral FET

Since the isolation of monolayer graphene by Novoselov & Geim et al.,^[163] graphene-based transistors have been of great interest for electronic applications. The primary advantage of graphene is its extremely high carrier mobility, which is theoretically estimated to be above $2 \times 10^5 \text{ cm}^2 \text{ V}^{-1} \text{ sec}^{-1}$ (Figure 4a).^[164] The mobility values of monolayer graphene FETs were reported to be about $1.8 \times 10^4 \text{ cm}^2 \text{ V}^{-1} \text{ sec}^{-1}$ on the SiO_2/Si back-gated structure^[11] and about $7.0 \times 10^4 \text{ cm}^2 \text{ V}^{-1} \text{ sec}^{-1}$ on the metal/ HfO_2 top-gated structure.^[12] Wang et al.^[13] measured the highest mobility of $1.4 \times 10^5 \text{ cm}^2 \text{ V}^{-1} \text{ sec}^{-1}$ in h -BN/graphene/ h -BN heterostructure-based FET with an edge-contact geometry; these results were attributed to suppressed carrier scattering and reduced contact resistance, as shown in Figure 4b. However, the lack of an energy bandgap in graphene results in an extremely low on/off-current ratio, which prevents graphene FETs from being used in logic applications. Alternatively, other 2D semiconducting materials, such as TMDs, have attracted much attention for use in 2D-material-based logic devices. These TMD materials have energy bandgap values of 1.0–1.8 eV; therefore, the FET devices fabricated on these TMDs exhibited a high on/off-current ratio of about 10^8 , as shown in Figure 4c.^[4] In Table 4, we provide a summary of the electronic device performances (mobility, on/off-current ratio, and subthreshold swing) of FETs formed on 2D materials including graphene and TMDs.^[4,11–13,20–33,62–64,67,68]

The electronic device performance of MoS_2 FETs was first estimated by Novoselov et al.,^[1] and the measured mobility values were very low (0.5 – $3 \text{ cm}^2 \text{ V}^{-1} \text{ sec}^{-1}$). Later, various fabrication techniques were developed and applied to improve the electronic device performance of 2D FETs. Du et al.^[20] enhanced the on-current and field-effect mobility of 2D FETs fabricated on MoS_2 using a PEI doping process. Electron carriers were

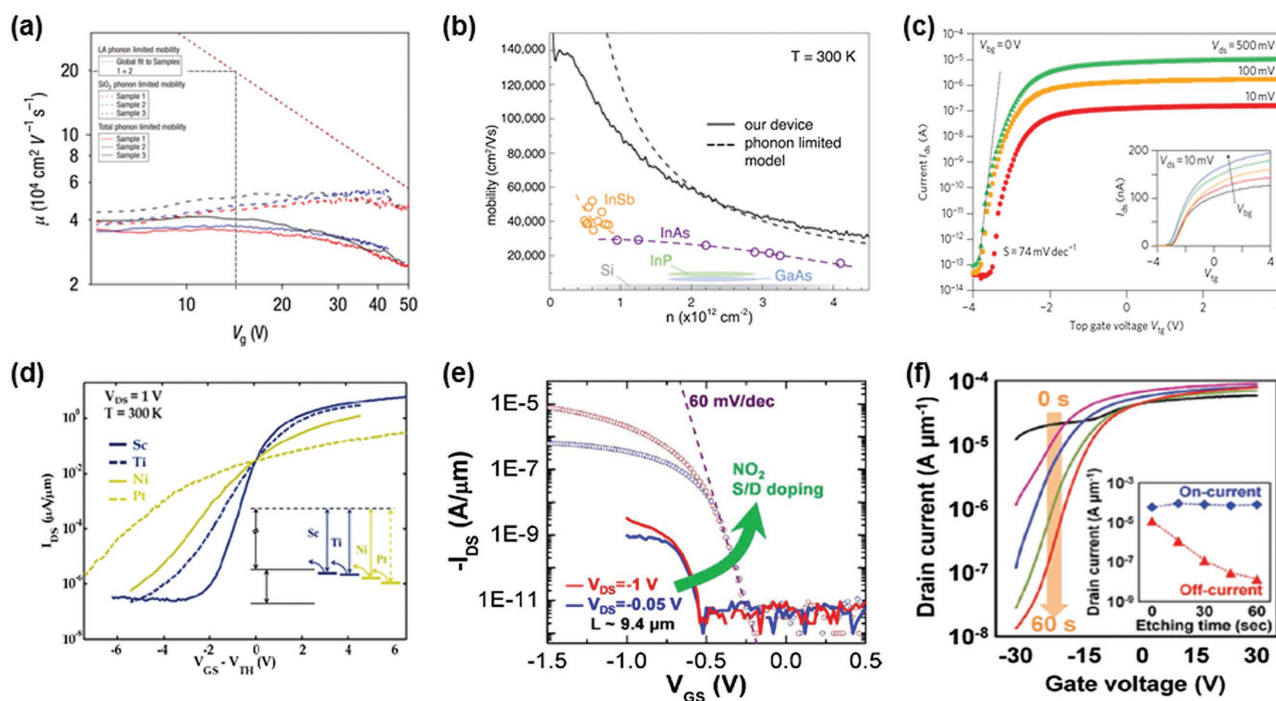


Figure 4. a) Mobility limits of graphene at room temperature (depending on gate voltage). Short-dashed, long-dashed, and solid lines correspond to scattering by acoustic phonons, SiO₂ surface phonons, and both phonon contributions, respectively. Reproduced with permission.^[164] Copyright 2008, Macmillan Publishers Limited. b) The graphene mobility at room temperature as a function of carrier density. Reproduced with permission.^[13] Copyright 2013, American Association for the Advancement of Science. c) I - V characteristics of monolayer MoS₂ transistor with HfO₂ gate dielectric. Reproduced with permission.^[4] Copyright 2011, Macmillan Publishers Limited. d) Electrical characteristics of four different metal (Sc, Ti, Ni, and Pt)/MoS₂ junction-based transistors. Reproduced with permission.^[21] Copyright 2013, American Chemical Society. e) I - V transfer curves of NO₂-doped monolayer WSe₂ transistor. Reproduced with permission.^[22] Copyright 2012, American Chemical Society. f) I - V characteristics of O₂ plasma-treated ReS₂ transistors as the O₂ plasma treatment time increases from 0 sec to 60 sec. Reproduced with permission.^[24]

transferred into the MoS₂ channel after applying the PEI doping technique, and the n-type concentration of MoS₂ increased, which reduced the contact and sheet resistances of the MoS₂ FET. Das et al.^[21] demonstrated a high-performance MoS₂ FET with a field-effect mobility of about $700 \text{ cm}^2 \text{ V}^{-1} \text{ sec}^{-1}$ by minimizing the Schottky barrier height between metal and MoS₂ using a scandium (Sc) electrode. As shown in Figure 4d, a very low effective barrier height (about 30 meV) was formed at the Sc/MoS₂ junction compared to the Pt/MoS₂ junction (about 230 meV) because the work function of Sc (3.5 eV) was lower than that of Pt (5.9 eV). Zhao et al. demonstrated a chemically assembled graphene/MoS₂/graphene FET, where the graphene was used as a metallic electrode.^[62] Because graphene's Fermi level can be adjusted by an applied gate bias, graphene was able to reduce the contact resistance between graphene and MoS₂. This contact resistance was reduced down to $10 \text{ k}\Omega \mu\text{m}$ when a high positive gate bias was applied, subsequently exhibiting a high on/off-current ratio of about 10^6 and a high field-effect mobility of $17 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.

Another TMD material, WSe₂, shows p-type or ambipolar behavior and has been researched for use in 2D electronic devices that are complementary to the MoS₂ FET. However, the electronic performance (e.g., mobility and on/off-current ratio) of WSe₂-based FETs was slightly lower than that of MoS₂ FETs^[22] because of the relatively high Schottky barrier height formed between the metal and WSe₂ by Fermi level pinning

(which is expected to cause high contact resistance). Thus, many studies have recently concentrated on improving the electronic device performance of the WSe₂ FET. Fang et al.^[22] presented a high-performance WSe₂-based FET with a high hole mobility of $250 \text{ cm}^2 \text{ V}^{-1} \text{ sec}^{-1}$, a low subthreshold swing of 60 mV dec^{-1} , and a high on/off-current ratio of 10^6 at room temperature. These values are comparable to those of high-performance MoS₂ FETs^[4,20,21] achieved using a NO₂ doping process at the S/D contact regions (Figure 4e). Chuang et al.^[23] reported a WSe₂ FET with graphene contact electrodes, which exhibited a high field-effect mobility of $300 \text{ cm}^2 \text{ V}^{-1} \text{ sec}^{-1}$ and a high on/off-current ratio of 10^7 . This performance was achieved by adjusting the Schottky barrier height at the graphene/WSe₂ junction through top-gate biasing. Recently, a high-performance ReS₂-based FET was achieved by Shim et al.^[24] through the use of O₂ plasma treatment. After the O₂ plasma treatment on bulk ReS₂ FETs, the ReS₂ active channel region was thinned, and the off-current was better controlled using an applied gate bias. This strategy resulted in an enhanced on/off-current ratio in the ReS₂ FETs of 10^4 (after O₂ plasma treatment) relative to a value of 10^1 (before O₂ plasma treatment), as shown in Figure 4f. Xu et al. presented a HfS₂ FET with a very high on/off-current ratio of 10^7 .^[63] Its on/off-current ratio and field-effect mobility values were strongly dependent on the metal contact and HfS₂ thickness. The HfS₂ FET with a Au contact and channel thickness between 7 and 12 nm exhibited maximal on/off-current ratio

Table 4. Performance comparison of various 2D-material-based lateral FETs.

Material	Structure	Temperature [K]	Mobility [$\text{cm}^2 \text{V}^{-1} \text{sec}^{-1}$]	On/off ratio [A/A]	Subthreshold swing [mV dec^{-1}]	Ref.
Graphene	Back-gated FET ^{a)}	1.6	$<1.8 \times 10^4$	–	–	[11]
	Organic solvent/Gr ^{b)} , Back-gated FET	300	7.0×10^4	–	–	[12]
	h-BN/Gr/h-BN, Back-gated FET	300	1.4×10^5	–	–	[13]
MoS ₂	Top-gated FET	300	2.2×10^2	10^8	74	[4]
	Back-gated FET, Molecular doping	–	3.3×10^1	10^5	–	[20]
	Back-gated FET	300	7.0×10^2	10^8	–	[21]
	Top-gated FET	300	1.1×10^3	10^8	178	[25]
	Dual-gated FET	300	5.2×10^2	10^8	–	[26]
	Top-gated FET	300	3.8×10^2	10^6	150	[27]
	Gr/MoS ₂ /Gr, Top-gated FET	300	1.7×10^1	10^6	–	[62]
WSe ₂	Top-gated FET, NO ₂ doping	300	2.5×10^2	10^6	60	[22]
	Ionic liquid gate FET	170	2.0×10^2	10^7	–	[23]
	Back-gated FET	300	3.0×10^2	10^6	250	[28]
	Ionic liquid gate FET	5	2.7×10^2	–	–	[29]
	Back-gated FET, air heating	300	9.2×10^1	10^8	–	[30]
ReS ₂	Back-gated FET, O ₂ plasma treatment	300	1.1×10^1	10^5	–	[24]
	Top-gated FET	300	1.0×10^0	10^6	750	[31]
	Dual-gated FET	77	2.6×10^1	10^5	148	[32]
	Back-gated FET	300	1.6×10^1	10^7	100	[33]
HfS ₂	Back-gated FET	300	2.4×10^0	10^7	–	[63]
GaTe	Back-gated FET	77	1.0×10^{-1}	10^5	–	[64]

^{a)}FET, field-effect transistor; ^{b)}Gr, graphene.

and field-effect mobility values of about 10^7 and $2.4 \text{ cm}^2 \text{V}^{-1} \text{s}^{-1}$, respectively. In addition, III–VI compounds 2D semiconductors such as GaTe have been studied.^[64] Recently, Wang et al. reported low off-current (10^{-12} A) and high on/off-current ratio (10^5) values in a GaTe FET by suppressing thermally activated Ga vacancy defects at liquid nitrogen temperature (77 K).^[64]

3.1.2. Vertical FET

Vertical heterojunction devices consisting of graphene and other 2D materials with a finite energy bandgap were suggested to resolve the low on/off-current ratio of graphene FETs and to achieve an extremely low operating voltage, which is difficult to accomplish in lateral TMD FETs (Table 5). Britnell et al.^[34] demonstrated a vertical FET based on a graphene/h-BN/graphene heterojunction. This FET was mainly operated by a direct tunneling mechanism (Figure 5a) that depends on the tunneling density of states in graphene and the effective barrier height between graphene and h-BN. The Fermi level in graphene was modulated by an applied gate voltage, affecting the tunneling density of states and the effective barrier height. However, this graphene/h-BN/graphene vertical FET exhibited a low on-current density (about $2 \times 10^{-5} \text{ A cm}^{-2}$) and a low on/off-current ratio (about 50). Yu et al.^[35] also presented a graphene/MoS₂ vertical FET with a high on/off-current ratio of about 10^3 and

a high on-current density of 5000 A cm^{-2} (Figure 5b). This improvement in on/off-current ratio and on-current density was attributed to the very low Schottky barrier height (150 meV) in graphene/MoS₂ heterojunction where thermionic emission phenomenon dominated the source–drain current. They also demonstrated a complementary inverter by sequentially stacking graphene, Bi₂Sr₂Co₂O₈, graphene, MoS₂, and metal, where the graphene/Bi₂Sr₂Co₂O₈/graphene and the graphene/MoS₂/metal vertical FETs worked as p- and n-channel devices, respectively. Similarly, many research groups have demonstrated graphene/MoS₂ heterojunction-based vertical FETs.^[36–39] In particular, Choi et al. achieved a high on-current density of $3.0 \times 10^3 \text{ A cm}^{-2}$ and an on/off-current ratio of 10^4 in a graphene/MoS₂ vertical FET by using an ion-gel gate dielectric.^[68] The high specific capacitance of the ion-gel-dielectric allowed for much wider modulation in the graphene's work function (0.43 eV) compared to the case of a 300-nm-thick SiO₂ gate dielectric (0.19 eV). Additionally, a graphene/WS₂/graphene vertical FET was reported by Georgiou et al. (Figure 5c).^[40] This FET was operated by both thermionic emission and tunneling mechanisms, and this device showed a high current modulation exceeding 10^6 at room temperature. Recently, Shim et al.^[41] demonstrated a graphene/WSe₂ vertical FET with p-channel characteristics, where a new carrier transport mechanism based on trap-assisted tunneling phenomenon was reported (Figure 5d). Because of this trap-assisted tunneling mechanism, the

Table 5. Performance comparison of various 2D-material-based vertical FETs.

Structure	Temperature [K]	On/off ratio [A/A]	On-current density [A cm ⁻²]	Off-current density [A cm ⁻²]	Operation voltage [V]	Ref.
Gr ^a /h-BN/Gr	240	5.0×10^1	$<1.5 \times 10^{-2}$	$<7.0 \times 10^{-4}$	0.2	[34]
Gr/MoS ₂ /Metal	150	1.5×10^4	1.5×10^1	1.0×10^{-3}	0.1	[35]
Gr/MoS ₂ /Metal	160	2.0×10^5	1.0×10^2	5.0×10^{-4}	0.1	[36]
Gr/MoS ₂ /Metal	300	1.0×10^5	1.0×10^4	1.0×10^{-1}	0.5	[37]
Gr/MoS ₂ /Metal	160	5.0×10^4	5.0×10^1	1.0×10^{-4}	0.1	[38]
Gr/MoS ₂ /Metal	90	3.5×10^1	1.3×10^{-1}	4.0×10^{-3}	0.05	[39]
Gr/MoS ₂ /Metal	300	1.0×10^4	$>3.0 \times 10^3$	$>3.0 \times 10^{-1}$	0.5	[68]
Gr/WS ₂ /Gr	300	1.0×10^6	$<2.0 \times 10^2$	$<2.0 \times 10^{-3}$	0.2	[40]
Gr/WSe ₂ /Metal	180	5.0×10^7	2.2×10^1	4.4×10^{-6}	0.1	[41]
Gr/WSe ₂ /Metal	300	1.0×10^4	$>3.1 \times 10^3$	$>3.1 \times 10^{-1}$	0.5	[68]
Gr/BP ^b /Metal	30	800	$<1.6 \times 10^3$	$<2.0 \times 10^1$	0.2	[67]

^a)Gr, graphene; ^b)BP, black phosphorus.

graphene/WSe₂ vertical FET exhibited an unexpected increase in on-current with a decrease in measurement temperature and an extremely high on/off-current ratio of 5×10^7 at 180 K, as shown in Figure 5e. Kang et al. demonstrated a graphene/BP vertical FET with two charge transport mechanisms, where thermionic emission and tunneling mechanisms were respectively dominant under high temperature + positive gate voltage and low temperature + negative gate voltage.^[67] In this case, due to the smaller energy bandgap and higher doping level in BP compared to other 2D materials, a low barrier height and

narrow barrier width were formed, leading to a tunneling current via defect and trap states in the bandgap region of BP.

3.2. Negative Differential Resistance (NDR) Device

The N-shaped NDR characteristic could be employed for future low-power applications. This NDR characteristic provides multiple threshold voltage values; therefore, an NDR device can be used to realize MVL circuits that require a larger number of

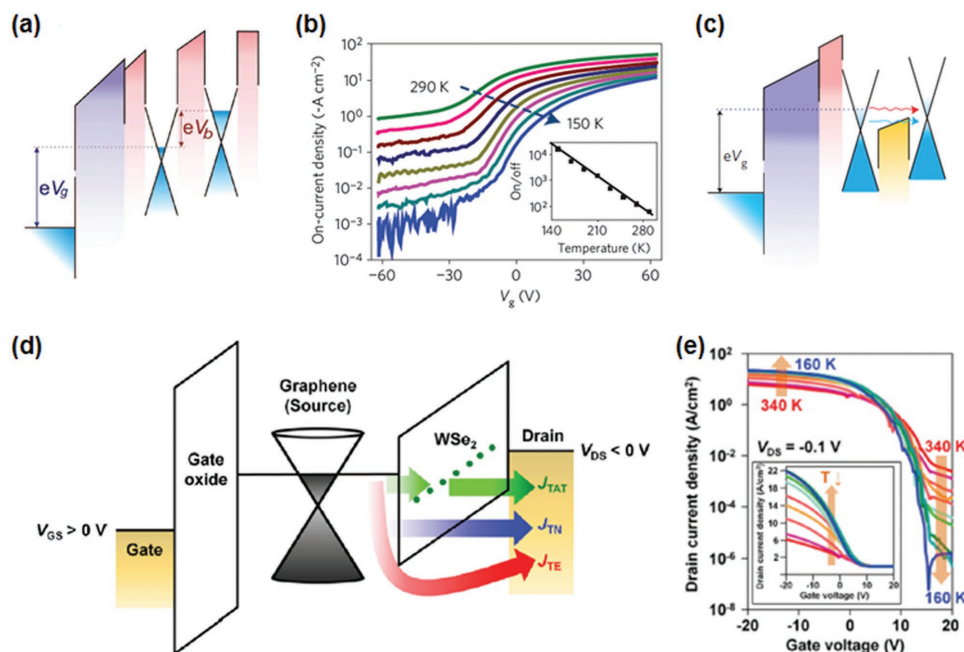


Figure 5. a) Energy band diagram of graphene/h-BN/graphene vertical FET. Reproduced with permission.^[34] Copyright 2012, American Association for the Advancement of Science. b) *I*-*V* characteristics of graphene/MoS₂ vertical FET at different temperatures. Reproduced with permission.^[35] Copyright 2012, Macmillan Publishers Limited. c) Energy band diagram of graphene/WS₂/graphene vertical FET under positive gate bias. The red and blue arrows indicate thermionic emission and direct tunneling currents, respectively. Reproduced with permission.^[40] Copyright 2012, Macmillan Publishers Limited. d) Energy band diagram of graphene/WSe₂/graphene vertical FET. e) *I*-*V* characteristics of the graphene/WSe₂/graphene vertical FET. Reproduced with permission.^[41]

logic states compared to traditional binary logic circuits. This MVL circuit can transfer more information with fewer interconnects, so it is expected to reduce the size and complexity of integrated circuits. Recently, 2D heterojunction-based NDR devices have been studied as 2D materials with very low surface defect states in principle allow high-quality heterojunctions with various band alignments (types I, II, and III) regardless of lattice mismatch issues. These 2D heterojunction-based NDR devices are classified as i) resonant tunneling diodes (RTDs)^[42–47] and ii) Esaki diodes^[48–50] (Table 6).

3.2.1. Resonant Tunneling Diodes (RTDs)

Britnell et al.^[42] reported an RTD device based on a graphene/*h*-BN/graphene heterostructure, where carriers were predicted to tunnel through a thin *h*-BN barrier layer when the electronic spectra (Dirac points) of the two graphene electrodes were aligned. In addition, an applied gate voltage changed the chemical potential of the bottom graphene layer, providing additional modulation of the effective barrier height and transmission coefficient. The 2D heterojunction-based RTD is predicted to show ultra-fast transit times compared to conventional RTDs made of quantum wells sandwiched between two potential barriers. This is because the RTD consists of a single tunnel barrier and the device operating speed is not limited by the dwell time of the carriers in the quantum well. Mishchenko et al.^[43] also observed NDR behavior in a graphene/*h*-BN/graphene heterostructure by aligning the crystallographic orientation of the two graphene layers. This resonant tunneling current was controlled by the relative orientation difference (momentum difference) between the two graphene crystalline lattices (Figure 6a). The resonant tunneling-based NDR behavior was also present between two bilayer graphene layers separated by an *h*-BN dielectric.^[44] Because the straight edges of bilayer graphene were rotationally aligned during the fabrication, more obvious resonance tunneling was observed compared to

that in monolayer graphene-based RTD devices. Kang et al.^[45] demonstrated a bilayer graphene/*h*-BN/bilayer graphene heterostructure-based RTD device with multiple NDR characteristics (four NDR peaks). These characteristics were observed due to the alignment of the two sub-bands of the two bilayer graphene (Figure 6b). The same group showed one transistor latch and static random access memory (SRAM) operation by connecting the RTD device to a load resistor in series, as shown in Figure 6c. This NDR behavior was also observed in trilayer MoS₂ sandwiched by top/bottom electrodes with nanopore structures.^[46] In this report, the interaction between a MoS₂ layer and an underlying Al₂O₃/Si₃N₄ membrane was predicted to confine the electron wave function in the nanopore, deform the electronic structure of MoS₂, and subsequently form discrete energy levels in each MoS₂ layer. A resonant tunneling phenomenon appeared when the quantized energy levels were aligned, resulting in an NDR characteristic at temperatures below 30 K. The resonant tunneling between two spatially separated quantum states was also observed in devices based on MoS₂/WSe₂/graphene and WSe₂/MoSe₂/graphene heterostructures, which were directly synthesized on graphene layers.^[47]

3.2.2. Esaki Diodes

The NDR characteristic was also observed by interlayer band-to-band tunneling. Roy et al.^[48] observed this NDR behavior in MoS₂/WSe₂ heterojunction-based Esaki diodes, where a dual-gate architecture was used to control the band alignment of the vertical heterojunction by independently modulating the electrostatic potential of MoS₂ (bottom) and WSe₂ (top) layers (Figure 7a). At a low applied voltage, electron carriers were predicted to tunnel from the valence band of the WSe₂ layer to the conduction band of the MoS₂ layer. This tunneling current then decreased with further increases in the applied voltage due to the forbidden bandgap of MoS₂, which showed a negative conductance. Nourbakhsh et al.^[49] claimed that the NDR phenomenon was attributed to the lateral band-to-band tunneling at the edge of the MoS₂/WSe₂ heterojunction, where a small effective bandgap of 50 meV was predicted to be formed by the energy band bending. A ternary inverter with three logic levels was demonstrated as a building block for MVL by using the sign changing property in the transconductance of the heterojunction NDR diode (Figure 7b and c). An NDR device was also realized in a BP/SnSe₂ heterostructure with a thin insulator layer, which was formed by the inter-diffusion of P and Sn atoms.^[50] The interfacial insulator (tunnel barrier) layer prevented the diffusion of electron carriers from the SnSe₂ (n-type region) to the BP (p-type region), affecting the valley current, and consequently the prominent NDR behavior was observed at room temperature.

3.3. Memory

Nonvolatile memory devices based on 2D materials have attracted much attention due to their miniaturization, low power consumption, and reliable data storage (Table 7). Early on, graphene-based nonvolatile memory devices were studied because of the high

Table 6. Summary of 2D-material-based NDR devices.

Materials	Device type	PVCR ^{a)} [A/A]	Ref.
Monolayer graphene/ <i>h</i> -BN/ Monolayer graphene	RTD ^{b)}	1.65 at 300 K, 4 at 7 K	[42]
Monolayer graphene/ <i>h</i> -BN/ Monolayer graphene		1.46 at 300 K, 2 at 2 K	[43]
Bilayer graphene/ <i>h</i> -BN/Bilayer Graphene		1.81 at 300 K, 1.43 at 10 K	[44]
Bilayer graphene/ <i>h</i> -BN/Bilayer Graphene		1.64 at 300 K	[45]
Trilayer MoS ₂		1.09 at 3 K	[46]
MoS ₂ /WSe ₂ /Graphene		2.3 at 300 K	[47]
WSe ₂ /MoSe ₂ /Graphene		2.2 at 300 K	[47]
MoS ₂ /WSe ₂	Esaki diode	1.86 at 77 K	[48]
MoS ₂ /WSe ₂		1.6 at 300 K	[49]
Black phosphorus/SnSe ₂		1.8 at 300 K	[50]

^{a)}PVCR, peak-to-valley current ratio; ^{b)}RTD, resonant tunneling diode.

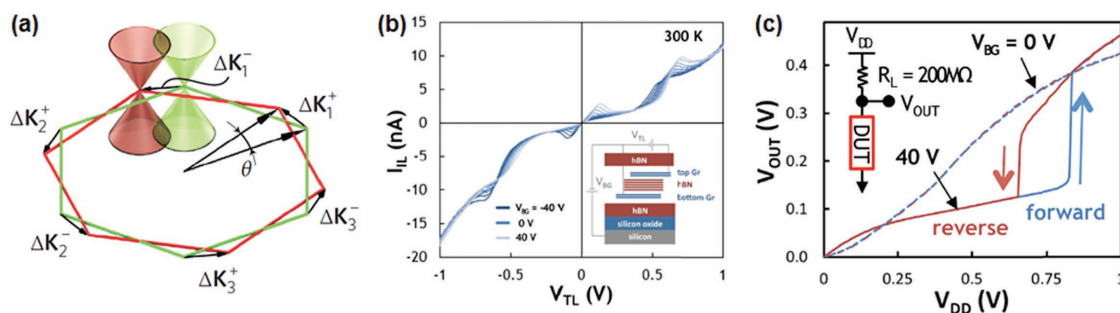


Figure 6. a) Energy band diagrams of graphene layers rotationally misaligned by θ between two Dirac points in real space. Reproduced with permission.^[43] Copyright 2014, Macmillan Publishers Limited. b) I - V curves of bilayer graphene/h-BN/bilayer graphene heterostructure-based RTD device showing multiple NDR characteristics. The inset shows a schematic of the RTD device formed on bilayer graphene/h-BN/bilayer graphene heterostructure. c) Output characteristics of a SRAM that was formed by connecting a load resistor (200 M Ω) to a bilayer graphene/h-BN/bilayer graphene heterostructure-based RTD device. Reproduced with permission.^[45] Copyright 2015, IEEE.

carrier mobility and mechanical flexibility of graphene.^[51–53] However, graphene memory devices typically exhibited very low program/erase (P/E) current ratios between 2 and 6 due to the absence of an energy bandgap.^[51–53] To overcome this limitation in graphene-based memory devices, MoS₂ with a sufficient energy bandgap (1.88 eV) was proposed as a channel material for non-volatile memory devices.^[54–58,65] The MoS₂-based FETs showed a high on/off-current ratio of approximately 10⁸,^[4] and this has the potential to enhance the P/E current ratio in nonvolatile memory systems based on MoS₂ FETs. Lee et al.^[54] and Lipatov et al.^[55] demonstrated MoS₂-based ferroelectric memory devices, where poly(vinylidene fluoride-trifluoroethylene) (P(VDF-TrFE)) and Pb(Zr,Ti)O₃ (PZT) were used as ferroelectric layers. The program and erase states of the ferroelectric memory devices were determined by controlling the polarization of the ferroelectric material by adjusting the gate voltage (Figure 8a). Zhang et al.^[56] also showed a nonvolatile charge-trap memory device in which MoS₂, Al₂O₃, and HfO₂ were employed as the channel, tunnel barrier, and charge-trapping layer, respectively. As shown in Figure 8b, electron carriers can tunnel from the channel into the floating gate through the barrier layer when the gate voltage is swept in the positive direction. Then, the injected electron carriers are

accumulated in the floating gate, and this induces a positive shift in the threshold voltage (program state). Conversely, when a gate voltage is swept in a negative direction, the electron carriers are transferred back from the floating gate to the channel, leading to a negative shift in the threshold voltage (erase state).

In addition, several researchers have demonstrated non-volatile memory devices that combine a graphene layer with a monolayer of MoS₂. Bertolazzi et al.^[57] reported a nonvolatile memory device based on a MoS₂/graphene heterostructure in which MoS₂ served as a channel layer. This MoS₂ channel was separated from a multilayer graphene (MLG) layer by a HfO₂ tunneling layer for charge trapping. This MLG allowed more charges to be stored in the floating gate because of its higher density of states ($>4.4 \times 10^{13} \text{ cm}^{-2} \text{ eV}^{-1}$) compared to that of monolayer graphene ($8 \times 10^{12} \text{ cm}^{-2} \text{ eV}^{-1}$); the overall result was a larger memory window.^[57] Similarly, Choi et al.^[58] presented a memory device consisting entirely of stacked 2D materials (graphene/h-BN/MoS₂). A larger hysteresis, a higher on/off-current ratio, and a high P/E current ratio were obtained in the stacked 2D memory devices by reducing the thicknesses of the h-BN tunneling barrier layer and graphene charge-trapping layer. Vu et al. demonstrated two-terminal tunneling random

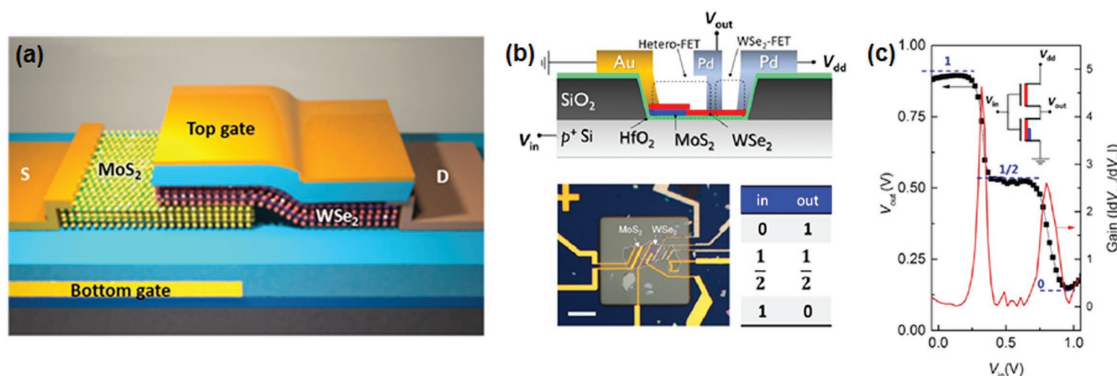


Figure 7. a) Three-dimensional schematic of a MoS₂/WSe₂ heterojunction-based NDR device with dual-gate architecture. Reproduced with permission.^[48] Copyright 2015, American Chemical Society. b) Schematic (top) and an optical image (down) of a ternary inverter with an input-output table of the inverter, where the ternary inverter is composed of a MoS₂/WSe₂ NDR device and a WSe₂ FET. c) Voltage transfer characteristic and gain of the ternary inverter formed with the MoS₂/WSe₂ NDR device. The inset shows a circuit schematic of the ternary inverter. Reproduced with permission.^[49] Copyright 2016, American Chemical Society.

Table 7. Summary of nonvolatile memory devices based on 2D materials.

Material	P/E ^{a)} current ratio [A/A]	Endurance [cycles]	Retention time [s]	Operating voltage [mV]	Ref.
Graphene	4	–	–	–	[51]
	2.55	100	1000	500	[52]
	5.5	1000	1000	–	[53]
MoS ₂	5,000	–	1000	1000	[54]
	10,000	500	10 000	100	[55]
	10,000	120	2000	50	[56]
Graphene, MoS ₂	10,000	120	2000	50	[57]
Graphene, <i>h</i> -BN, MoS ₂	100,000	110	1400	50	[58]
Graphene, <i>h</i> -BN, MoS ₂	100,000	100 000	10 000	8000	[65]
BP ^{b)}	100	120	1200	300	[59]
	1,000	–	1000	100	[60]
MoS ₂ , <i>h</i> -BN, BP	50	40	1000	50	[61]

^{a)}P/E, program/erase; ^{b)}BP, black phosphorus.

access memory with a significantly low off-state current of 10^{-14} A and a very high P/E-current ratio over 10^5 by vertically stacking MoS₂, *h*-BN and graphene.^[65] Furthermore, they confirmed excellent stretchable ability by successfully fabricating two-terminal MoS₂/*h*-BN/graphene memory devices on polyimide (PI)/polydimethylsiloxane (PDMS) with no significant degradation in electrical performance. The tunneling process

for programming and erasing originated from the asymmetric potential drop at two contacts of source/floating gate (graphene) and the drain/floating gate. A nonvolatile charge-trap memory device on BP material was also recently reported.^[59–61] Because FETs fabricated on BP exhibit ambipolar characteristics, nonvolatile memory devices with a BP channel are able to store both electron and hole carriers in the charge trapping layer. A type-switching memory device formed with a BP/*h*-BN/MoS₂ heterostructure was developed based on this property of BP devices, and the memory device operated under p- or n-type modes according to the polarity of a gate voltage pulse (Figure 8c).^[61] In addition, Lee et al.^[60] showed a memory inverter circuit that directly converts analog signals to digital signals by connecting the BP (channel)/Al₂O₃ (tunnel barrier)/BP (charge-trapping) heterostructure-based memory device to an external load resistor (Figure 8d). These 2D material (MoS₂ and BP)-based memory devices exhibited high-performance in terms of P/E current ratio (10^2 – 10^5), retention characteristics (30–40% loss/10 years), and operating voltage (0.05–0.3V).^[55–61] Overall, it is expected that nonvolatile memory devices based on 2D materials will provide a promising route for miniaturization and power-saving for portable electronic devices requiring memories embedded in system-on-a-chip devices.

4. Application of 2D materials to Optoelectronics

Graphene has received much interest because of its exceptional optical and electrical characteristics. Due to unique electronic band structure of graphene, it can absorb 2.3% of incident light^[165] and generate a photocurrent through photothermoelectric effects (the temperature difference by light illumination is converted to an electric voltage).^[166] In addition, the huge carrier mobility (about 200000 cm² V⁻¹ s⁻¹) of graphene can be exploited in ultra-fast 2D-material-based optoelectronic applications. A monolayer of MoS₂ was used in optoelectronic devices by Mak et al.^[167] as a new 2D semiconductor material with a wide direct bandgap ($E_g = 1.88$ eV). This allowed high light absorption over a wide spectral range (visible to near-infrared). Other TMDs (e.g., MoSe₂, WS₂, and WSe₂) have also been reported to have a direct bandgap in the monolayer structure.^[168] This discovery of graphene and TMD could open the door for next-generation 2D-material-based optoelectronic applications. In this section, we review the development of 2D-material-based optoelectronic devices: i) photodetectors,^[3,5,14,15,24,31,69–83] ii) photovoltaic devices,^[84–95] and iii) LEDs.^[87,88,95–101]

4.1. Photodetectors

After the discovery of the unique optical properties of monolayer graphene, many

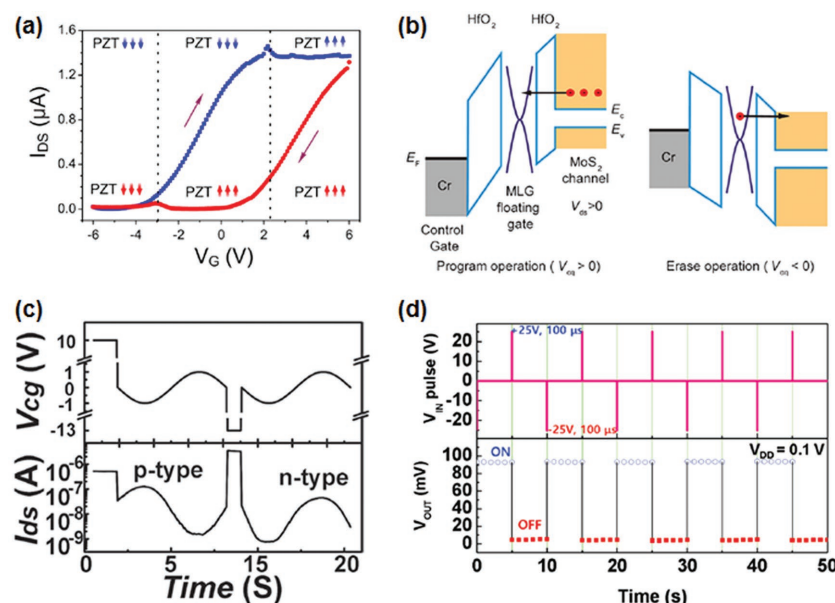


Figure 8. a) I_D - V_G characteristics of a MoS₂/PZT ferroelectric memory device. The arrows indicate the polarization directions of PZT at different gate voltages. Reproduced with permission.^[55] Copyright 2015, American Chemical Society. b) Band diagrams explaining the program and erase states in a charge-trap memory device under positive and negative control gate voltages (V_{cg}), respectively. Reproduced with permission.^[57] Copyright 2013, American Chemical Society. c) Electrical characteristics of a type-switching memory device formed on a BP/*h*-BN/MoS₂ heterostructure. The memory device worked under p-type mode after a 10 V pulse was applied, and it was switched to n-type mode after a -13 V pulse was applied. Reproduced with permission.^[61] d) Dynamic output voltage (V_{OUT}) response of a BP-based memory inverter. Reproduced with permission.^[60]

studies focused on graphene-based photodetectors. Xia et al.^[14] presented an ultra-fast graphene photodetector with a maximum responsivity of 0.5 mA W^{-1} under a 1550 nm wavelength laser and a very high bandwidth ($>500 \text{ GHz}$). Gan et al.^[15] also reported a chip-integrated graphene photodetector with a responsivity of 15.7 mA W^{-1} under a 2000 nm wavelength laser. However, various problems caused by the lack of a bandgap in graphene (e.g., high dark current, low light absorption, and low quantum yield) prevented the successful development of high-performance 2D photodetectors. These problems suggest that another 2D material should be used in these applications. Mak et al.^[167] discovered that monolayer MoS_2 (group VI TMD material) has an electronic band structure with a direct bandgap, and they confirmed the strong photon-electron interactions. Since then, MoS_2 has received attention as a candidate for high-performance 2D photodetectors requiring low dark-current and high responsivity.^[69] Yin et al.^[5] and Lopez-Sanchez et al.^[3] also demonstrated monolayer MoS_2 -based photodetectors with responsivity values of 7.5 mA W^{-1} and 880 A W^{-1} , respectively. The photothermoelectric effect at the metal/ MoS_2 junction enables the absorption of light with a lower energy than the optical energy bandgap of MoS_2 ; as such, MoS_2 photodetectors are capable of showing a wide spectral response from the UV to the near-infrared.^[69] The previously mentioned optical properties (direct bandgap, strong photon-electron interaction, and photothermoelectric effect) were also found in other TMDs (e.g., MoSe_2 , WS_2 , and WSe_2), thereby accelerating the development

of TMD-based photodetectors.^[93,169] Pradhan et al.^[70] fabricated trilayer WSe_2 photodetectors with high responsivity of about 7 A W^{-1} under white illumination. The highest responsivity value of $2.5 \times 10^7 \text{ A W}^{-1}$ was reported in a ReS_2 phototransistor by Shim et al.^[24]

Although the responsivity values of the TMD-based photodetectors were much higher than those of monolayer graphene photodetectors (about 15.7 mA W^{-1}),^[15] the slow switching speed of the TMD photodetectors is a significant problem that should be resolved for furtherance of the TMD devices (about 50 ms on MoS_2 ^[69] and $<50 \text{ ps}$ on graphene^[71]). In addition, atomically thin 2D semiconductors have physical limitations associated with light absorption in spite of their very high absorption coefficient (exceeding $5 \times 10^7 \text{ m}^{-1}$ in the visible range^[170]) compared to 3D semiconductors with sufficient film thickness.^[171] To ameliorate the performance of 2D-material-based photodetectors (e.g., responsivity, detectivity, and switching speed), recent studies have concentrated on the development of various techniques including i) chemical and plasma treatments on the device channel, ii) Schottky barrier height modulation between metal and 2D material, iii) formation of heterojunctions, and iv) usage of an absorption layer on the device channel. We summarized the performance of photodetectors based on graphene and TMDs according to the technique, as shown in Table 8.

One of the most promising strategies for enhancing 2D-material-based optoelectronic device performance is to perform chemical or plasma treatment on the device channel. For

Table 8. Comparison of the performance in various 2D-material-based photodetectors.

Method	Material	Treatment/structure	Wavelength [nm]	Responsivity [A W^{-1}]	Detectivity [Jones]	Switching speed [s]	Ref.
Single structure	Gr ^{a)}	–	1550	5.0×10^{-4}	–	–	[14]
		–	2000	1.6×10^{-2}	–	–	[15]
	MoS_2	–	550	7.5×10^{-3}	–	5×10^{-3}	[5]
		–	561	8.8×10^2	–	4–9	[3]
		–	white	7.0×10^0	–	4×10^{-5}	[70]
Chemical/plasma treatment	MoS_2	PPh_3 ^{b)}	520	6.13×10^5	–	8.31	[73]
		OTS ^{c)} /APTES ^{d)}	655	5.7×10^3	4.5×10^9	–	[74]
	WSe_2	PPh_3	520	1.27×10^6	–	2.8×10^{-3}	[72]
		OTS/APTES	655	1.5×10^4	5.3×10^{10}	–	[74]
	ReS_2	O_2 plasma	405	2.5×10^7	–	6.7×10^{-1}	[24]
	ReSe_2	PPh_3	520	3.68×10^4	–	3×10^{-2}	[73]
	WSe_2	Ti-, Pd- contact	650	1.8×10^5	10^{14}	>10	[75]
Contact region modulation		Ti-, Pt-contact	520	3.39×10^3	–	3×10^{-2}	[72]
Heterojunction	WSe_2	Gr/ WSe_2 /Gr	759	2.0×10^{-2}	–	5.5×10^{-12}	[77]
	$\text{WSe}_2/\text{MoS}_2$	$\text{WSe}_2/\text{Gr}/\text{MoS}_2$	635	1.0×10^4	10^{15}	5.36×10^{-5}	[78]
Inserting absorption layer	MoS_2	Perovskite	520	1.94×10^6	1.29×10^{12}	6.2	[79]
		PbS	635	6.0×10^5	5.0×10^{11}	3.5×10^{-1}	[80]
		R6G ^{f)}	520	1.17×10^0	1.5×10^7	2.3	[81]

^{a)}Gr, graphene; ^{b)} PPh_3 , triphenylphosphine; ^{c)}OTS, octadecyltrichlorosilane; ^{d)}APTES, aminopropyltriethoxysilane; ^{e)}Perov, perovskite; ^{f)}R6G, rhodamin 6G.

example, an electron doping process can improve the responsivity of a MoS₂ photodetector because the recombination rate of photocarriers is suppressed by tightly-bounded trions formed in the MoS₂ channel.^[172] Furthermore, Jo et al.^[72,73] achieved high photoresponsivity in MoS₂ (1.18×10^6 A W⁻¹), WSe₂ (1.27×10^6 A W⁻¹), and ReSe₂ (6.13×10^5 A W⁻¹) photodetectors by reducing the contact resistance and blocking the charge puddle effect through PPh₃ and aminopropyltriethoxysilane (APTES) treatment techniques. These treatments increased the collection of photocarriers and reduced the scattering effect at the TMD/SiO₂ interface, respectively (Figure 9a and b). Kang et al.^[74] applied octadecyltrichlorosilane (OTS) and APTES treatment techniques onto WSe₂ and MoS₂ photodetectors and performed photoluminescence analysis. They adjusted the amount of tightly-bounded trions in MoS₂ and positively-charged trions in WSe₂ by changing the electron concentration in TMD layers via the OTS and APTES techniques. This resulted in improvements in the responsivity of the MoS₂ photodetector, but decreased the responsivity of the WSe₂ photodetector. In addition, a plasma treatment processes also improved the optoelectronic performance of TMD photodetectors. Shim et al.^[24] performed O₂ plasma treatment on ReS₂ phototransistors, as shown in Figure 9c. The plasma treatment increased the defect density in the ReS₂ channel, and the created defects eventually improved the switching speed of the photodetector by

shortening photocarrier lifetime. After the O₂ plasma treatment was performed on the ReS₂ surface for 60 sec, the rising and decaying time values were improved from 16.7 sec to 0.67 sec and from 25.2 to 5.6 sec, respectively (Figure 9d). The switching speed of the 2D-based photodetector was also improved by adjusting the Schottky barrier height between the metal electrode and the 2D material. When the photodetector device was exposed to light, photocarriers generated in the depletion region separated toward the metal electrodes due to an electric field. Here, the photoresponse time is dependent on the magnitude of the electric field formed at the metal/semiconductor junction. This junction can be adjusted by changing the metal work function; as a result, the photocarriers can be more rapidly separated by the strong electric field.^[173] Zhang et al.^[75] showed fast photoresponse times in a Ti/WSe₂ junction-based photodetector (23 ms) compared to a Pd/WSe₂ junction photodetector (5 s) by achieving a high Schottky barrier height and a strong electric field at the Ti/WSe₂ junction using a low-work function metal (Ti) (Figure 10a and b). Similarly, Jo et al.^[72] also reported an enhancement in photoresponse time (from about 420 ms to 200 ms) in a WSe₂ photodetector by changing the contact metal from Pt to Ti.

Although TMD-based photodetectors showed fast switching speeds (about 40 μ s), they are still slow compared to graphene-based photodetectors (about 50 ps). In order to overcome this limitation in switching speed, many researchers recently concentrated on the fabrication of van der Waals vertical heterostructure-based photodetectors by stacking graphene and TMD layers. The vertically stacked monolayer TMD device had an atomically short channel length to provide ultra-fast charge transfer between electrodes.^[76] Massicotte et al.^[77] reported *h*-BN/graphene/WSe₂/graphene/*h*-BN heterostructure-based photodetectors with a fast photoresponse time of 5.5 ps (Figure 10c), which is comparable to that of graphene photodetectors (<50 ps).^[71] These photodetectors based on 2D heterojunctions also showed wide spectral detection from the visible to the far infrared. Long et al.^[78] demonstrated a photodetector with a WSe₂/graphene/MoS₂ heterostructure, which was able to absorb light in an extremely wide spectral range between 400 nm and 2400 nm (Figure 10d). Photocarriers were mainly generated in WSe₂ and MoS₂ when a laser with a short wavelength (<750 nm) was exposed to this photodetector because the photon energy was higher than the energy bandgaps of WSe₂ and MoS₂. Meanwhile, long-wavelength light (between 750 nm to 2400 nm) was absorbed in the graphene layer, and the generated photocarriers were transferred from the graphene to the TMD layers by an electric field.

The use of a light absorption layer on 2D materials was also proposed as another approach to improve responsivity, detection, and photodetection range of the 2D

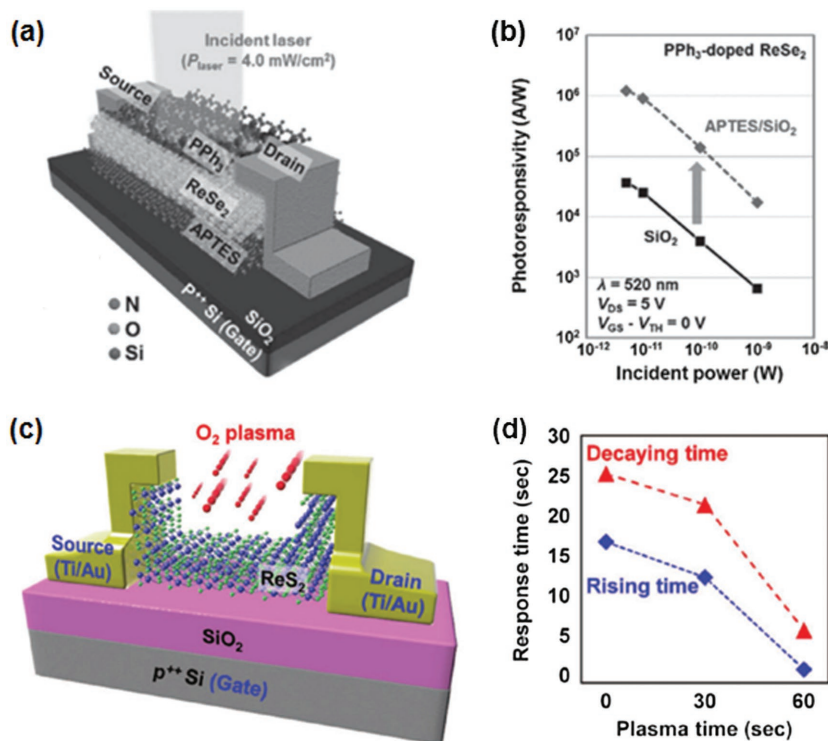


Figure 9. a) Three-dimensional schematic illustration of a PPh₃/ReSe₂/APTES photodetector. b) Responsivity of the PPh₃-doped ReSe₂ on SiO₂ and APTES/SiO₂ substrates, where $V_{\text{DS}} = 5 \text{ V}$ and $\lambda = 520 \text{ nm}$. Reproduced with permission.^[73] c) Three-dimensional illustration of an O₂-plasma-treated ReS₂ photodetector. d) Responsivity of the O₂-plasma-treated ReS₂ photodetector for 0 to 60 sec, where $\lambda = 405 \text{ nm}$ and $V_{\text{GS}} = 30 \text{ V}$. Reproduced with permission.^[24]

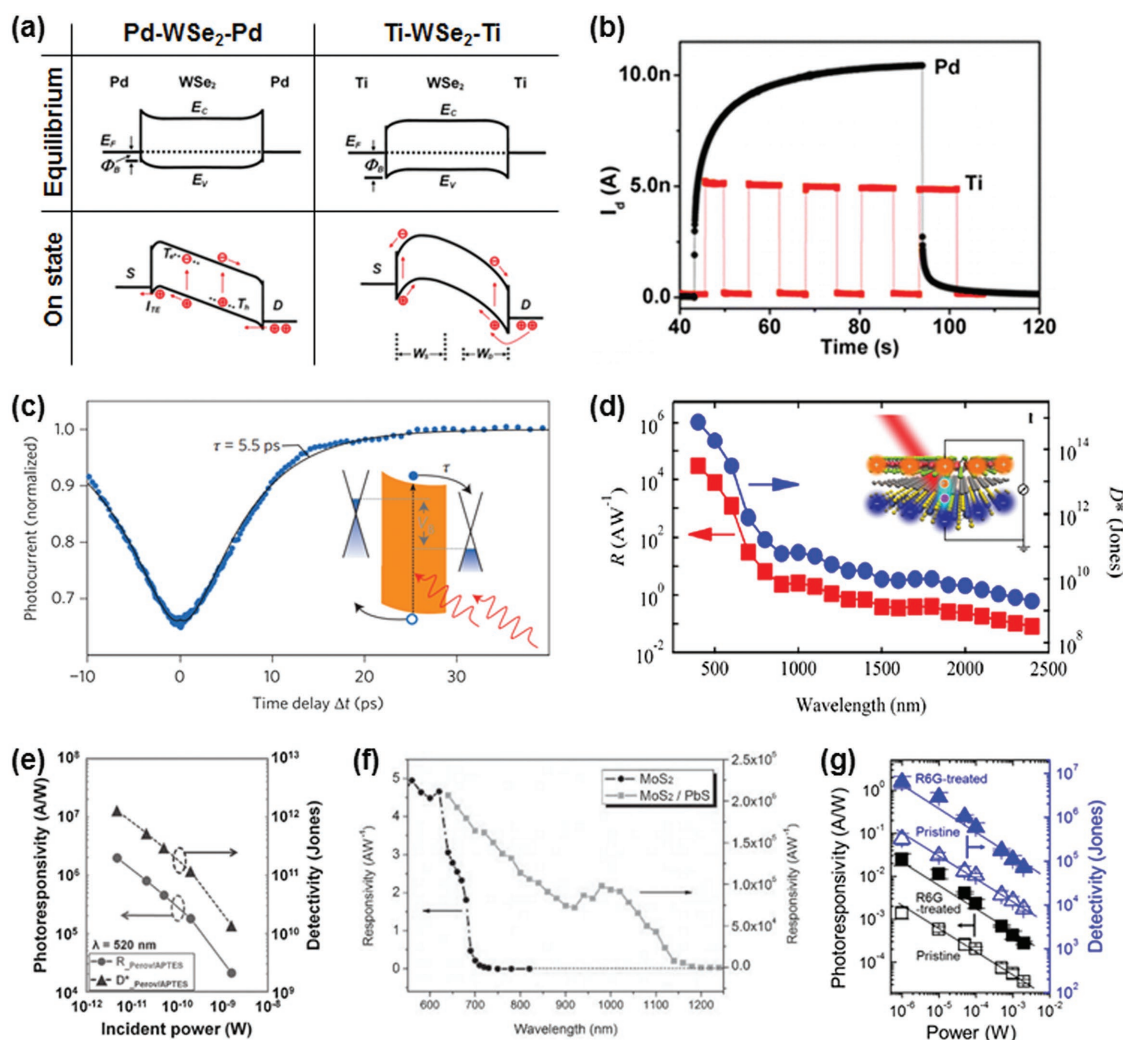


Figure 10. a) Energy band diagrams of a WSe₂ photodetector with Pd and Ti metal contacts at equilibrium and in the on-state. b) Photoresponse time characteristics of Pd and Ti/WSe₂ junction-based photodetectors. Reproduced with permission.^[75] Copyright 2014, American Chemical Society. c) Temporal response of a graphene/WSe₂/graphene heterostructure-based photodetector. The inset shows the energy band diagram of the graphene/WSe₂/graphene heterostructure under laser exposure. Reproduced with permission.^[77] Copyright 2016, Macmillan Publishers Limited. d) Responsivity of a WSe₂/graphene/MoS₂ photodetector as a function of wavelength. The inset shows a three-dimensional illustration of the WSe₂/graphene/MoS₂ heterostructure under laser exposure. Reproduced with permission.^[78] Copyright 2014, American Chemical Society. e) Responsivity and detectivity values of a perovskite/MoS₂/APTES photodetector as a function of laser incident power. Reproduced with permission.^[79] f) Responsivity of pristine MoS₂ and PbS/MoS₂ photodetectors as a function of exposed laser wavelength. Reproduced with permission.^[80] g) Responsivity and detectivity values of pristine MoS₂ and R6G-treated MoS₂ photodetectors as a function of laser incident power. Reproduced with permission.^[81] Copyright 2014, American Chemical Society.

photodetectors. A number of photocarriers can be generated in the device by depositing an absorption layer with very high absorption rate and quantum efficiency. In this strategy, the carriers are expected to transfer from the absorption layer to the channel layer, thereby enhancing photocurrent. Kang et al.^[79] recently presented an ultra-high-performance photodetector based on a CH₃NH₃PbI₃ (perovskite)/TMD hybrid structure. The responsivity and detectivity values were dramatically improved relative to a control due to the outstanding optical properties of perovskite (e.g., high absorption rate, ambipolar charge transport, and long diffusion length). The responsivity improved from $8.16 \times 10^2 \text{ A W}^{-1}$ to $2.12 \times 10^4 \text{ A W}^{-1}$, and the detectivity improved from $3.93 \times 10^9 \text{ Jones}$ to $1.38 \times 10^{10} \text{ Jones}$

(Figure 10e). PbS quantum dot films also helped improve light absorption on a MoS₂-based photodetector.^[80] As shown in Figure 10f, PbS quantum dots absorbed light in a wide spectral range from 400 nm to 1500 nm to supplement the light absorption in the infrared region. The detection range of the MoS₂ photodetector is typically below 680 nm. Similarly, Yu et al.^[81] also demonstrated a dye-sensitized MoS₂ photodetector with enhanced spectral photoresponse by using a rhodamine 6G (R6G) absorption layer. The detection response of the R6G/MoS₂ photodetector was extended to the infrared region ($\lambda < 980 \text{ nm}$) because the R6G has a very wide optical bandgap (about 2.38 eV). In addition, the photocarriers generated in the R6G layer were well transferred to the MoS₂

channel, thereby enhancing the responsivity up to 1.17 A W^{-1} (Figure 10g).

4.2. Photovoltaic Devices

2D materials can absorb 5–10% of incident sunlight over a thickness of less than 1 nm, which is similar to the absorbance of 20-nm-thick Si or 5-nm-thick GaAs.^[139,157] In spite of this excellent light absorption, the low built-in potential of the metal–2D-material–metal junction causes low open circuit voltage (V_{OC}) and low power conversion efficiency (PCE), which hinder the successful fabrication of practical photovoltaic devices. Thus, new approaches have been explored to obtain high built-in potentials inside photoactive 2D layers that can facilitate the dissociation of photocarriers and enhance the PCE value.^[84,85] Recently, many research groups have implemented 2D-material-based photovoltaic devices with a high built-in potential by forming an asymmetric i) Schottky junction^[84,86,92] or ii) PN junction^[85,87–91,93–95] inside the photoactive 2D layer, as summarized in Table 9. Shanmugam et al.^[84] demonstrated a photovoltaic device based on an asymmetric Schottky junction with Au and indium tin oxide (ITO) contacts. These contacts formed a high Schottky barrier and an Ohmic contact on MoS₂, respectively (Figure 11a). The large work function difference between Au and ITO generated a high electric field in the depletion region of MoS₂, and this efficiently dissociated the photocarriers toward ITO and Au contacts, eventually exhibiting a high PCE of 1.8% (Figure 11b). A gate-tunable photovoltaic device formed in a graphene/MoS₂/graphene heterostructure was also developed by Yu et al.^[86] (Figure 11c). The Schottky barrier height between graphene and MoS₂ can be modulated by an applied external electric field through the gate electrode because graphene has a finite density of states. When a negative gate voltage was applied, the work function of graphene increased, and the Schottky barrier height in the graphene/MoS₂ junction also increased. In addition, the band slope in MoS₂ became sharper

due to an increase in the built-in electric field. This consequently promoted the separation of photocarriers, resulting in a high open circuit voltage (V_{OC}) of 0.3 V and short circuit current (I_{SC}) of $2 \times 10^{-6} \text{ A}$.

Several research groups have developed PN-junction-based photovoltaic devices to achieve relatively high built-in potentials in active 2D layers regardless of the limitations of Fermi level pinning phenomenon between the metal and 2D semiconductor. Pospischil et al.^[87] and Baugher et al.^[88] reported photovoltaic devices based on a WSe₂ PN junction, where laterally separated double gate electrodes were used to define the PN junction (Figure 11d). In addition, Wi et al.^[85] demonstrated a MoS₂-based photovoltaic device with a high PCE of 2.8% by forming a PN junction through CHF₃ plasma treatment on one side of the MoS₂ (plasma-treated MoS₂: p-type and untreated MoS₂: n-type), as shown in Figure 11e. In this work, the CHF₃ plasma-induced PN junction was estimated to have a built-in potential of about 0.7 eV. Recently, this photovoltaic effect was also observed in hetero-PN-junctions, which can be easily formed by just stacking p- and n-type 2D materials (Figure 11f). Furchi et al.^[89] and Yi et al.^[90] showed photovoltaic devices fabricated on WSe₂/MoS₂ hetero-PN-junctions, which had PCE values of 0.2% and 1.5%, respectively. Furthermore, a GaTe/MoS₂ photovoltaic device with an EQE of 61.68% and a PCE of 0.45% was also reported by Wang et al.,^[91] where GaTe was an intrinsic p-type 2D material that exhibited a direct bandgap property irrespective of its thickness ($E_g = 1.7 \text{ eV}$ for multilayer).

4.3. Light-Emitting Diodes (LEDs)

Monolayer TMD materials, such as Mo- and W-based dichalcogenides, have direct energy bandgaps (1–2 eV) with strong photoluminescence (PL) characteristic, making them suitable for use in LEDs.^[6,174–176] Recently, many research groups have reported TMD-based 2D LEDs using the excellent electron-hole pair recombination property at various junctions including i) Schottky junctions,^[96–98] ii) PN junctions,^[87,88,95,99] and

Table 9. Summary of 2D-material-based photovoltaic devices.

Materials	Junction type	Light source	$I_{SC}^a)$	$V_{OC}^b)$ [V]	FF ^{c)}	PCE ^{d)} [%]	Ref.
ITO/MoS ₂ /Au	Schottky	AM 1.5G	5.37 mA cm^{-2}	0.59	0.55	1.8	[84]
Gr ^{e)} /MoS ₂ /Gr		514 nm	$2 \times 10^{-6} \text{ A}$	0.3	–	–	[86]
Pd/MoS ₂ /Au		532 nm	9.7 mA cm^{-2}	0.1	0.37	0.36	[92]
Au/MoS ₂ /ITO	PN	AM 1.5G	20.9 mA cm^{-2}	0.28	0.47	2.8	[85]
Pd/WSe ₂ /Ti		Halogen lamp	1.6 mA cm^{-2}	0.85	0.5	0.5	[87]
Au/WSe ₂ /Au		700 nm	$1.7 \times 10^{-9} \text{ A}$	0.5	–	–	[88]
Pd/WSe ₂ /MoS ₂ /Pd		Halogen lamp	$5 \times 10^{-11} \text{ A}$	0.55	0.5	0.2	[89]
Cr/WSe ₂ /MoS ₂ /Pd		AM 1.5G	$2.5 \times 10^{-10} \text{ A}$	0.2	0.45	0.15	[90]
Cr/GaTe/MoS ₂ /Cr		473 nm	$5 \times 10^{-9} \text{ A}$	0.07	0.28	0.42	[91]
Cr/WSe ₂ /Cr		640 nm	$4.1 \times 10^{-10} \text{ A}$	0.7	–	0.14	[93]
Gr/WSe ₂ /MoS ₂ /Gr		532 nm	$7 \times 10^{-8} \text{ A}$	0.5	–	–	[94]
Ni/WSe ₂ /MoS ₂ /Au		514 nm	$2.2 \times 10^{-7} \text{ A}$	0.27	–	–	[95]

^{a)} I_{SC} , short circuit current; ^{b)} V_{OC} , open circuit voltage; ^{c)} FF, fill factor; ^{d)} PCE, power conversion efficiency; ^{e)} Gr, graphene.

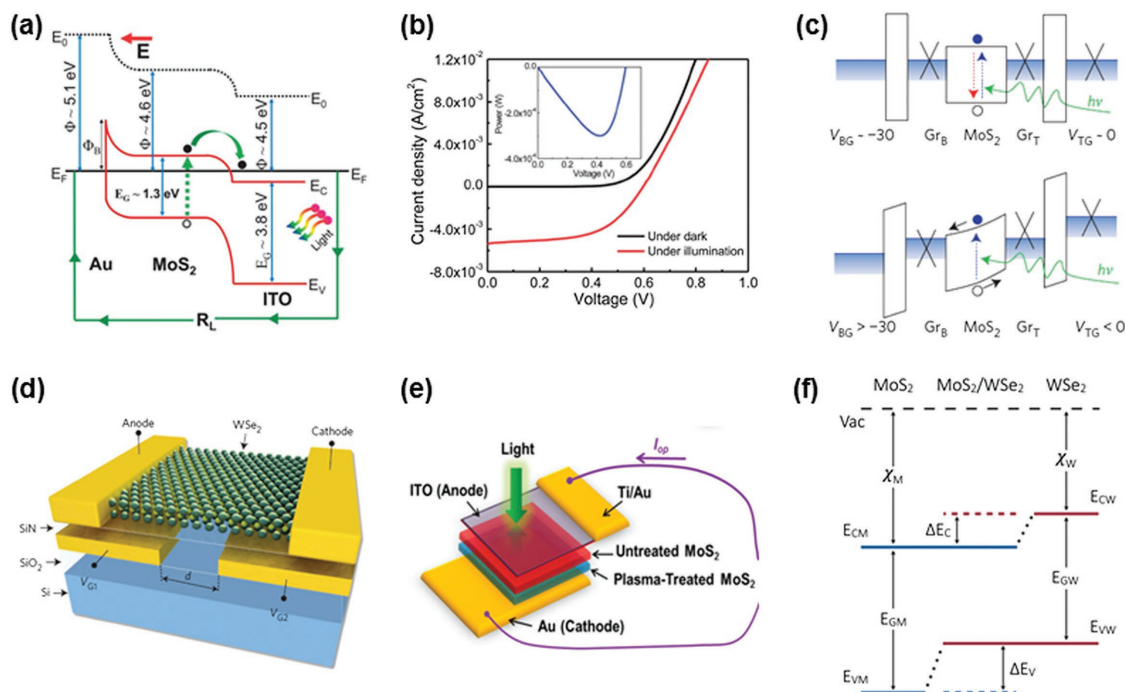


Figure 11. a) Energy band diagram of Au/MoS₂/ITO explaining the charge transfer in a Schottky junction-based photovoltaic device. b) Measured dark and photo *J*-*V* characteristics of the Au/MoS₂/ITO Schottky junction-based photovoltaic device. Reproduced with permission.^[84] Copyright 2012, The Royal Society of Chemistry. c) Schematic band diagrams of a photovoltaic device based on dual-gated graphene/MoS₂/graphene vertical heterostructure. Reproduced with permission.^[86] Copyright 2013, Macmillan Publishers Limited. d) Schematic of laterally separated double gate structure for WSe₂ PN junction-based photovoltaic device. Reproduced with permission.^[87] Copyright 2014, Macmillan Publishers Limited. e) Schematic of a CHF₃-plasma-treated MoS₂ photovoltaic device. Reproduced with permission.^[85] Copyright 2014, American Chemical Society. f) Energy band diagram of a MoS₂/WSe₂ hetero-PN-junction. Reproduced with permission.^[89] Copyright 2014, American Chemical Society.

iii) tunnel junctions,^[100,101,177] which are detailed in Table 10. Sundaram et al.^[96] first observed the EL phenomenon in MoS₂ devices with Schottky junctions formed between MoS₂ and Cr contacts, as shown in Figure 12a. During the application of a high drain voltage to the MoS₂ device, carriers injected from the Cr to the MoS₂ were accelerated in a strong electric

field, eventually creating many electron-hole pairs by impact ionization.^[96] A high EL peak was then observed at the MoS₂/Cr Schottky junction, where distinct light emission exists only above a threshold power density of 15 kW cm⁻². This light emission phenomenon was also demonstrated in other TMDs (WS₂^[97] and WSe₂^[98]) using Schottky junctions between TMDs

and metal contacts. Several research groups reported EL phenomena based on radiative recombination at PN junctions, as shown in Figure 12b.^[87,88,95,99] This light emission process was more efficient than that of the Schottky junction, which was dependent on electron-hole pair generation by impact ionization.^[96] Highly efficient light emission (EL efficiency of 0.1–1%) was demonstrated at low currents (from 35 to 200 nA) in WSe₂ by using a PN junction electrostatically doped by the dual gates.^[87,88,99] In addition, an EL spectrum in a vertically stacked MoS₂/WSe₂ heterojunction (n-type for MoS₂ and p-type for WSe₂) was also observed by Cheng et al.^[95] (Figure 12c). However, light emission at a chemically doped PN junction has not yet been reported. Withers et al.^[100,101] recently reported tunneling LEDs fabricated on vertically stacked heterojunctions consisting of metallic graphene (electrode), insulating *h*-BN (tunnel

Table 10. Various LED devices fabricated on 2D materials.

Type	Material	Structure	EL ^{a)} peak position [nm]	Ref.
Schottky junction	MoS ₂	Cr/MoS ₂ /Cr	685	[96]
	WS ₂	Au/WS ₂ /Au	630	[97]
	WSe ₂	Ti/WSe ₂ /Ti	740	[98]
p-n junction	WSe ₂	Au/n-WSe ₂ /p-WSe ₂ /Au	750	[99]
		Au/n-WSe ₂ /p-WSe ₂ /Pd	752	[88]
		Ti/n-WSe ₂ /p-WSe ₂ /Pd	802	[87]
	MoS ₂ /WSe ₂	Ni/n-MoS ₂ /p-WSe ₂ /Au	810	[95]
	MoSe ₂	Gr/ <i>h</i> -BN/MoSe ₂ / <i>h</i> -BN/Gr	670	[100]
Tunnel junction	MoSe ₂	Gr/ <i>h</i> -BN/MoSe ₂ / <i>h</i> -BN/Gr	770	[101]
	WS ₂	Gr/ <i>h</i> -BN/WS ₂ / <i>h</i> -BN/Gr	630	[100]
	WSe ₂	Gr/ <i>h</i> -BN/WSe ₂ / <i>h</i> -BN/Gr	760	[101]
	Gr	Gr/ <i>h</i> -BN/WSe ₂ / <i>h</i> -BN/Gr	727	[177]
	WSe ₂	Cr/ <i>h</i> -BN/n-WSe ₂ /p-WSe ₂ / <i>h</i> -BN/Gr	728	[177]

^{a)}EL, electroluminescence; ^{b)}Gr, graphene.

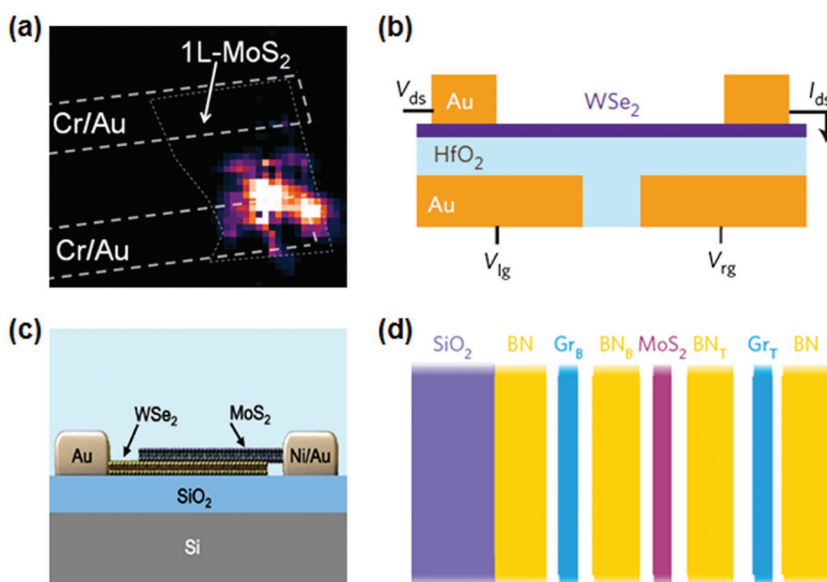


Figure 12. a) False color image showing EL emission in the MoS₂ with Cr/Au contacts. Reproduced with permission.^[96] Copyright 2013, American Chemical Society. b) Schematic diagram of the WSe₂ device controlled by two local gates. Reproduced with permission.^[88] Copyright 2014, Nature Publishing Group. c) Schematic illustration of the WSe₂/MoS₂ vertical heterojunction device. Reproduced with permission.^[95] Copyright 2014, American Chemical Society. d) Schematic of the single quantum well heterostructure (*h*-BN/Gr_B/*h*-BN/MoS₂/*h*-BN/Gr_T/*h*-BN). Reproduced with permission.^[100] Copyright 2015, Nature Publishing Group.

barrier), and semiconducting TMD (light emitter) layers (Figure 12d). The carriers injected from the graphene electrode easily accumulated at the TMD emitter layer because the direct tunneling phenomenon was suppressed by *h*-BN tunnel barriers between the graphene electrode and TMD emitter layer. Consequently, the LED device based on this vertical tunnel junction showed more efficient light emission (EL efficiency of 5%) than that of the PN junction^[87,88,99] (EL efficiency of 0.1–1%). This EL efficiency in a tunneling LED was comparable to that used for modern organic LEDs.^[178] Additionally, Clark et al.^[177] reported a single photon emitter in a WSe₂ device using both vertical tunnel and lateral PN junctions. In this work, the vertical LED device consisted of two graphene transparent electrodes, two *h*-BN tunnel barriers, and one central WSe₂ layer (graphene/*h*-BN/WSe₂/*h*-BN/graphene). Two separated back gates were utilized for the lateral LED device to electrostatically define the PN junction in WSe₂. They observed a narrow EL peak^[177] (line width < 300 μ eV) at 1.704 eV (728 nm), and this emission line was similar to the PL peak^[179] previously reported in WSe₂ single photon emitters.

5. Conclusion

In this review, we introduced recent progress on the fabrication techniques for improving the performance of 2D-material-based devices, such as i) doping, ii) contact engineering, and iii) heterojunction formation. Next, we discussed 2D-material-based electronic and optoelectronic device applications,

including lateral/vertical FETs, NDR devices, memory devices, photodetectors, photovoltaic devices, and LEDs. The rediscovery of graphene has encouraged researchers to also have an interest in other atomically thin 2D materials, such as TMDs. Therefore, the TMD materials have been considered as alternative semiconductors for next-generation electronic and optoelectronic device applications with novel functionalities due to their thickness scalability at the atomic level, finite energy bandgap values (1.0–1.8 eV), strong photon–electron interaction, and high light absorption. In particular, not only single junction but also various heterojunction devices based on these 2D materials have afforded novel functionality and advances in many research fields. The availability of various 2D materials could allow the formation of a wide variety of heterojunctions with desired band alignments, and it thereby provided a powerful fabrication process platform for high-performance electronic and optoelectronic devices. We believe that the various research results related to 2D semiconductor fabrication process and device technologies will attract the researchers to be more interested in 2D-material-based device applications,

and this will eventually cause 2D materials to be utilized for practical high-performance electronic and optoelectronic applications in the future.

However, many studies related to 2D-material-based electronic and optoelectronic devices have been demonstrated at the micrometer-scale by using a mechanical exfoliation method. Although a single crystal graphene layer was grown at the wafer-scale (>2 inch),^[180] efforts to grow other 2D materials, such as WSe₂, ReS₂, and BP, are still in their infancy. In this light, it is essential to grow high quality 2D materials at the wafer-scale with large grain sizes and uniform film thicknesses to realize practical device applications based on 2D materials. Furthermore, because most reported 2D heterojunction devices were fabricated by using a mechanical dry transfer process that is too complicated and inefficient for practical applications, development of the process to directly synthesize 2D materials on another 2D material at a large scale would be one of the ultimate goals for successfully integrating the 2D heterojunction-based devices at the industrial level.

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